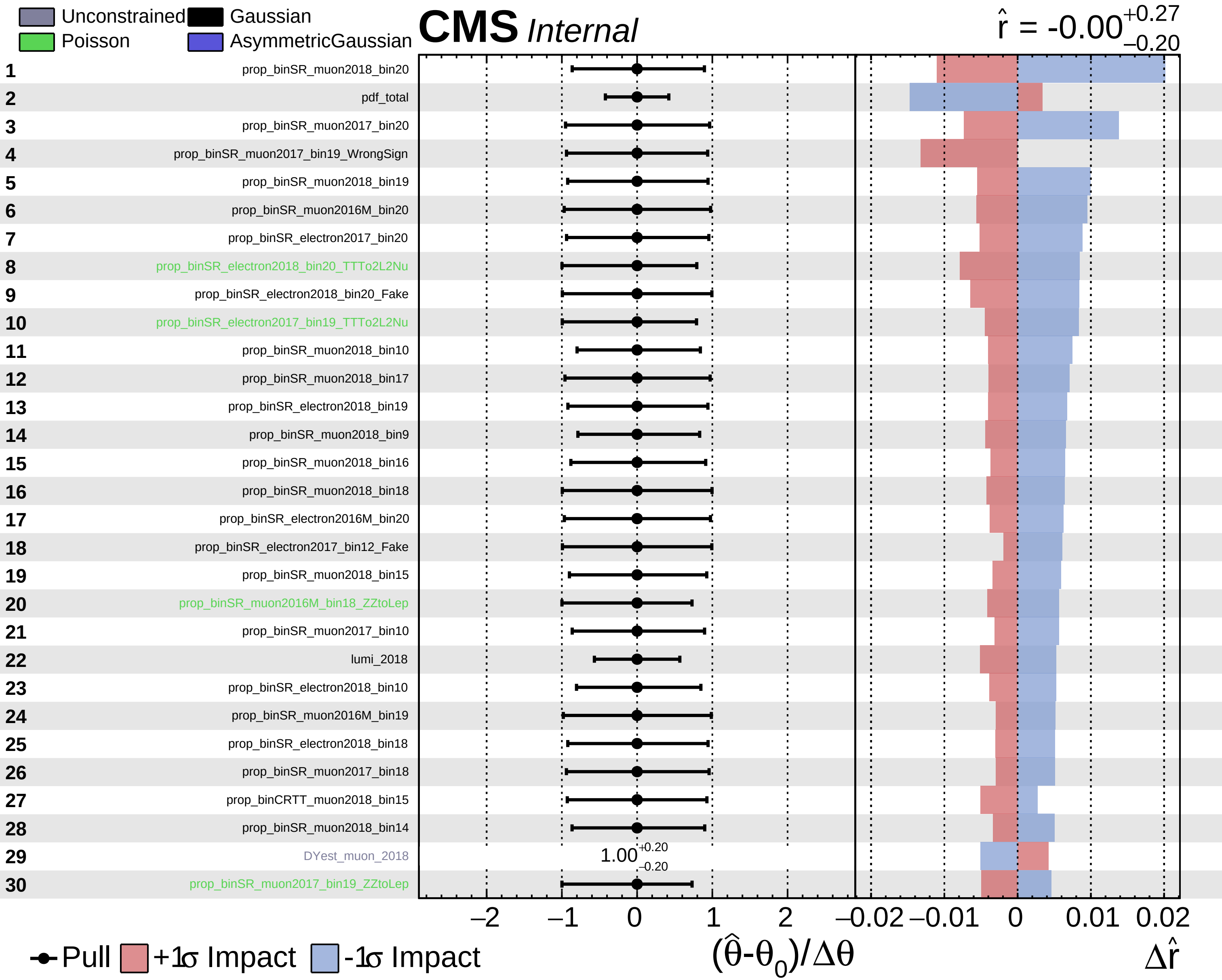
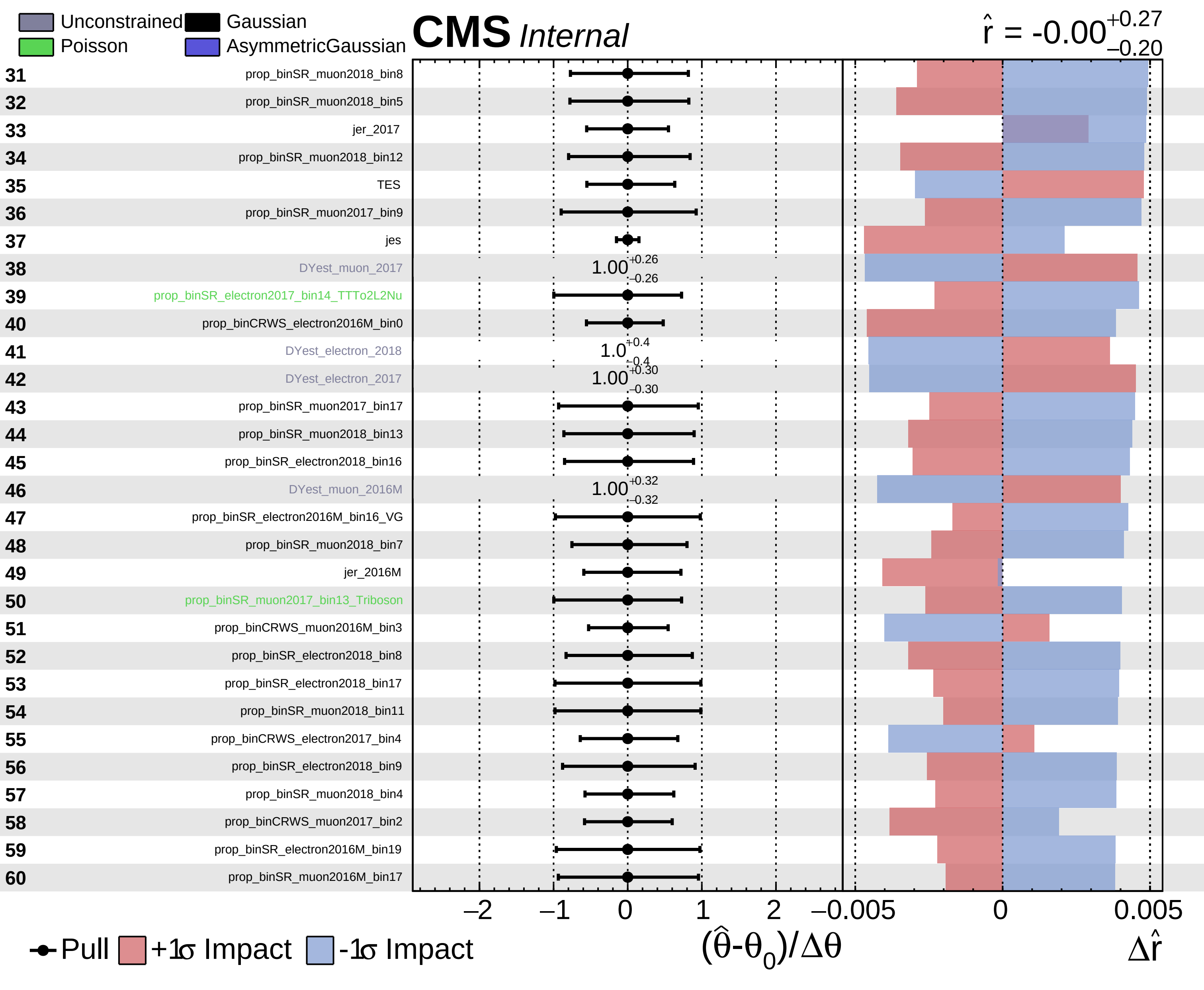
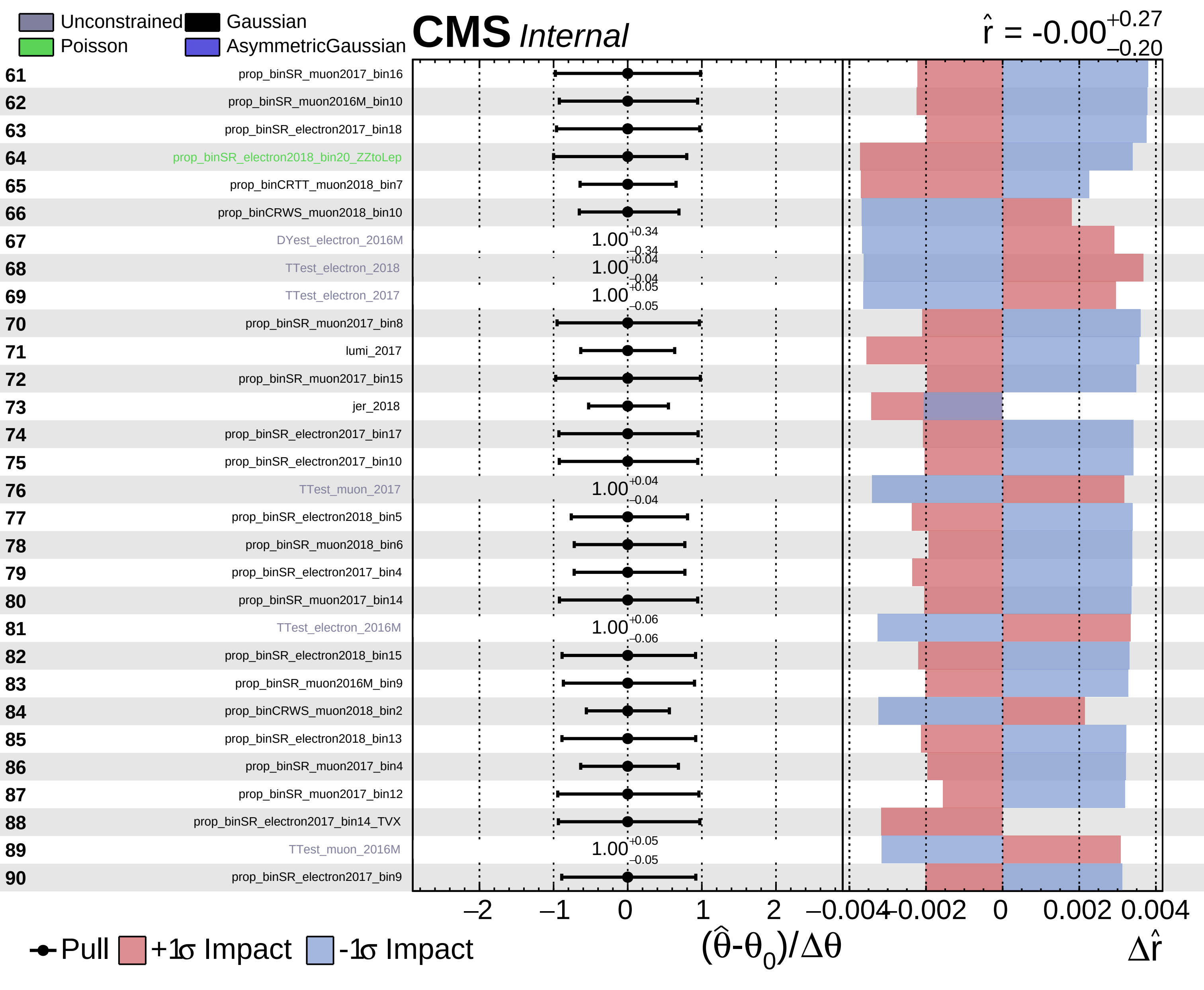


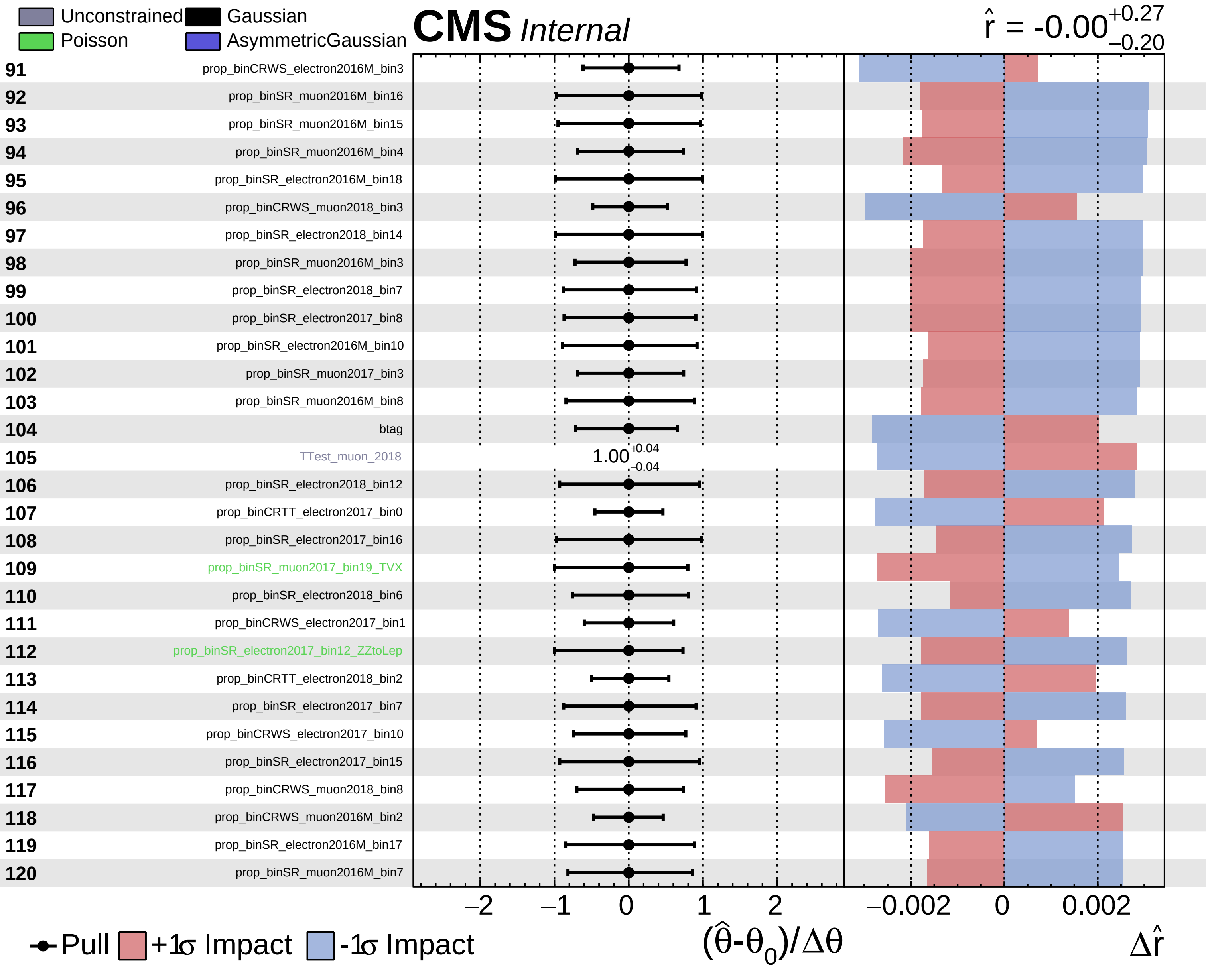
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
1 prop_binSR_muon2018_bin20
2 pdf_total
3 prop_binSR_muon2017_bin20
4 prop_binSR_muon2017_bin19_WrongSign
5 prop_binSR_muon2018_bin19
6 prop_binSR_muon2016M_bin20
7 prop_binSR_electron2017_bin20
8 prop_binSR_electron2018_bin20_TTTo2L2Nu
9 prop_binSR_electron2018_bin20_Fake
10 prop_binSR_electron2017_bin19_TTTo2L2Nu
11 prop_binSR_muon2018_bin10
12 prop_binSR_muon2018_bin17
13 prop_binSR_electron2018_bin19
14 prop_binSR_muon2018_bin9
15 prop_binSR_muon2018_bin16
16 prop_binSR_muon2018_bin18
17 prop_binSR_electron2016M_bin20
18 prop_binSR_electron2017_bin12_Fake
19 prop_binSR_muon2018_bin15
20 prop_binSR_muon2016M_bin18_ZZtoLep
21 prop_binSR_muon2017_bin10
22 lumi_2018
23 prop_binSR_electron2018_bin10
24 prop_binSR_muon2016M_bin19
25 prop_binSR_electron2018_bin18
26 prop_binSR_muon2017_bin18
27 prop_binCRTT_muon2018_bin15
28 prop_binSR_muon2018_bin14
29 DYest_muon_2018
$1.00^{+0.20}_{-0.20}$
30 prop_binSR_muon2017_bin19_ZZtoLep
−2
−1
0
1
2
−0.02
−0.01
0
0.01
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



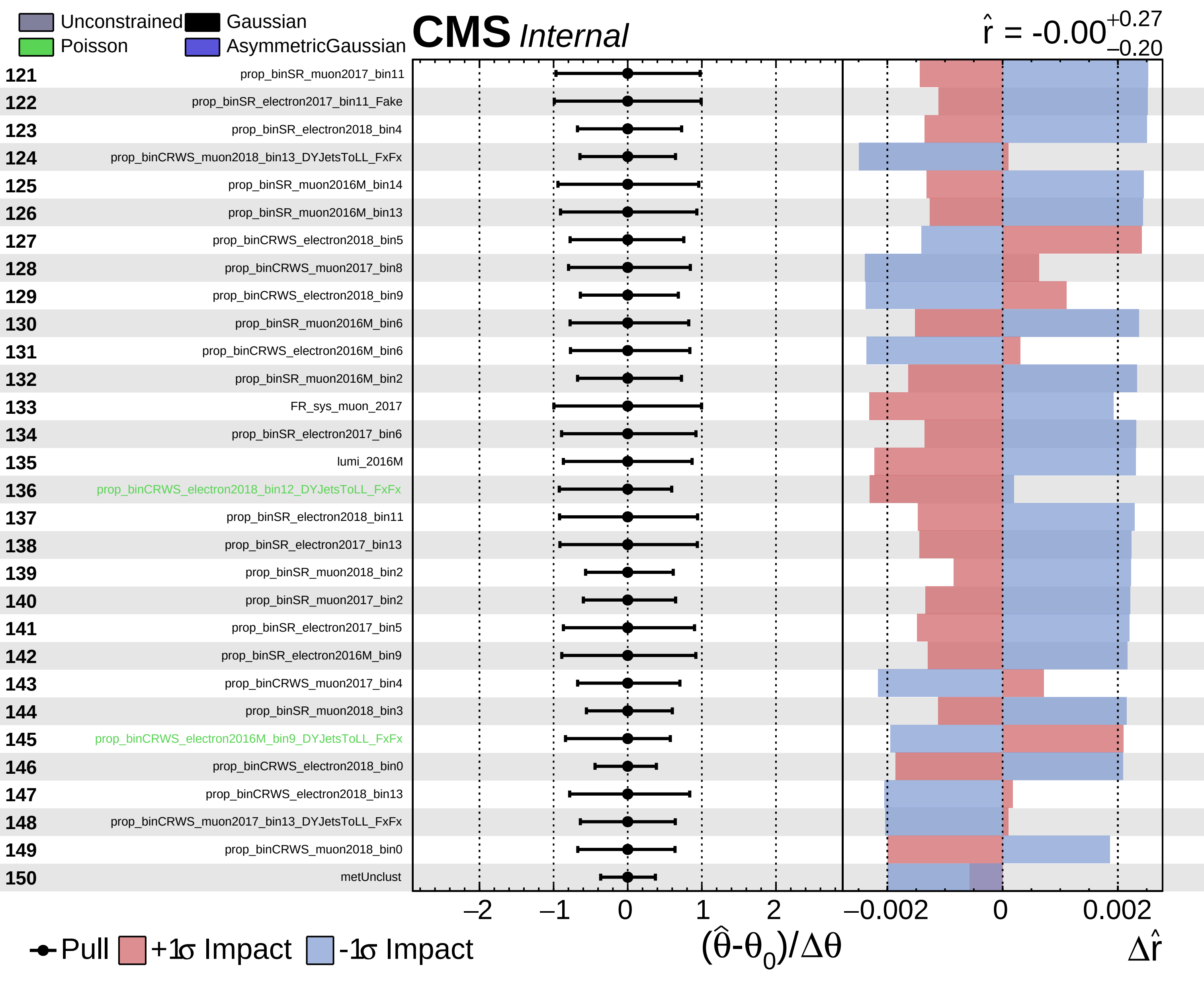
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
31 prop_binSR_muon2018_bin8
32 prop_binSR_muon2018_bin5
33 jer_2017
34 prop_binSR_muon2018_bin12
35 TES
36 prop_binSR_muon2017_bin9
37 jes
38 DYest_muon_2017 $1.00^{+0.26}_{-0.26}$
39 prop_binSR_electron2017_bin14_TTTo2L2Nu
40 prop_binCRWS_electron2016M_bin0
41 DYest_electron_2018 $1.0^{+0.4}_{-0.4}$
42 DYest_electron_2017 $1.00^{+0.30}_{-0.30}$
43 prop_binSR_muon2017_bin17
44 prop_binSR_muon2018_bin13
45 prop_binSR_electron2018_bin16
46 DYest_muon_2016M $1.00^{+0.32}_{-0.32}$
47 prop_binSR_electron2016M_bin16_VG
48 prop_binSR_muon2018_bin7
49 jer_2016M
50 prop_binSR_muon2017_bin13_Triboson
51 prop_binCRWS_muon2016M_bin3
52 prop_binSR_electron2018_bin8
53 prop_binSR_electron2018_bin17
54 prop_binSR_muon2018_bin11
55 prop_binCRWS_electron2017_bin4
56 prop_binSR_electron2018_bin9
57 prop_binSR_muon2018_bin4
58 prop_binCRWS_muon2017_bin2
59 prop_binSR_electron2016M_bin19
60 prop_binSR_muon2016M_bin17
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

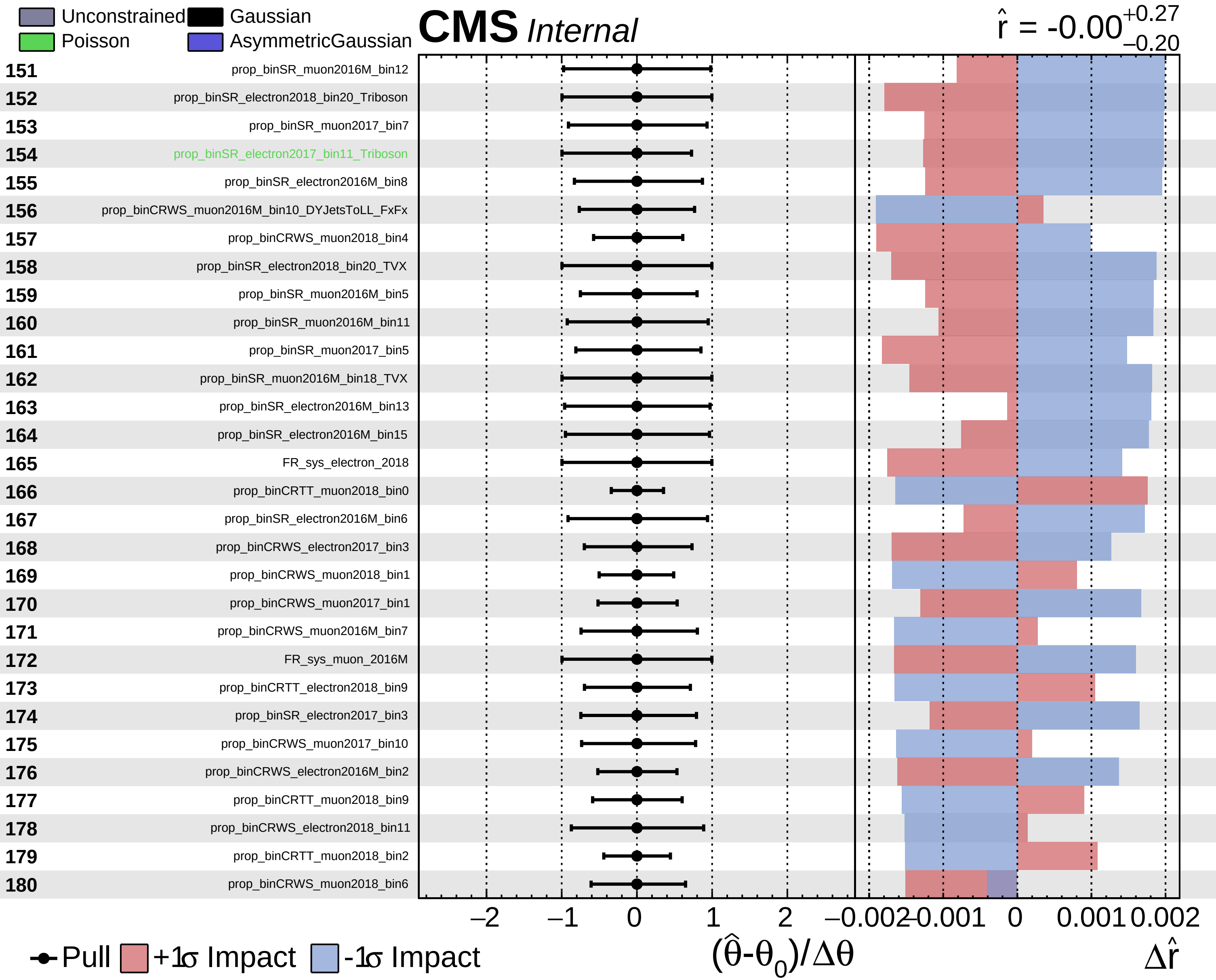


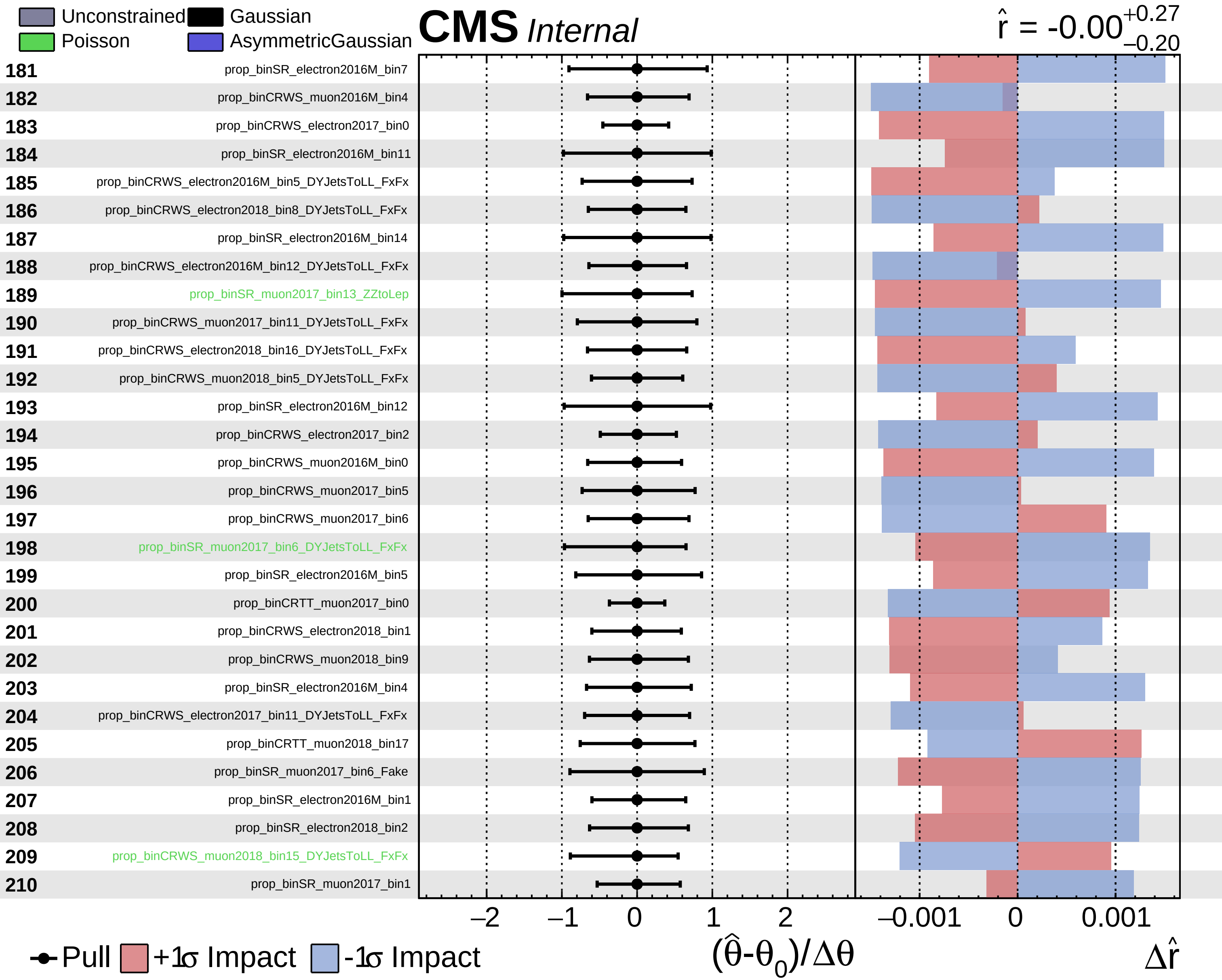
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
61 prop_binSR_muon2017_bin16
62 prop_binSR_muon2016M_bin10
63 prop_binSR_electron2017_bin18
64 prop_binSR_electron2018_bin20_ZZtoLep
65 prop_binCRTT_muon2018_bin7
66 prop_binCRWS_muon2018_bin10
67 DYest_electron_2016M
$1.00^{+0.34}_{-0.34}$
68 TTest_electron_2018
$1.00^{+0.04}_{-0.04}$
69 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
70 prop_binSR_muon2017_bin8
71 lumi_2017
72 prop_binSR_muon2017_bin15
73 jer_2018
74 prop_binSR_electron2017_bin17
75 prop_binSR_electron2017_bin10
76 TTest_muon_2017
$1.00^{+0.04}_{-0.04}$
77 prop_binSR_electron2018_bin5
78 prop_binSR_muon2018_bin6
79 prop_binSR_electron2017_bin4
80 prop_binSR_muon2017_bin14
81 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
82 prop_binSR_electron2018_bin15
83 prop_binSR_muon2016M_bin9
84 prop_binCRWS_muon2018_bin2
85 prop_binSR_electron2018_bin13
86 prop_binSR_muon2017_bin4
87 prop_binSR_muon2017_bin12
88 prop_binSR_electron2017_bin14_TVX
89 TTest_muon_2016M
$1.00^{+0.05}_{-0.05}$
90 prop_binSR_electron2017_bin9
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



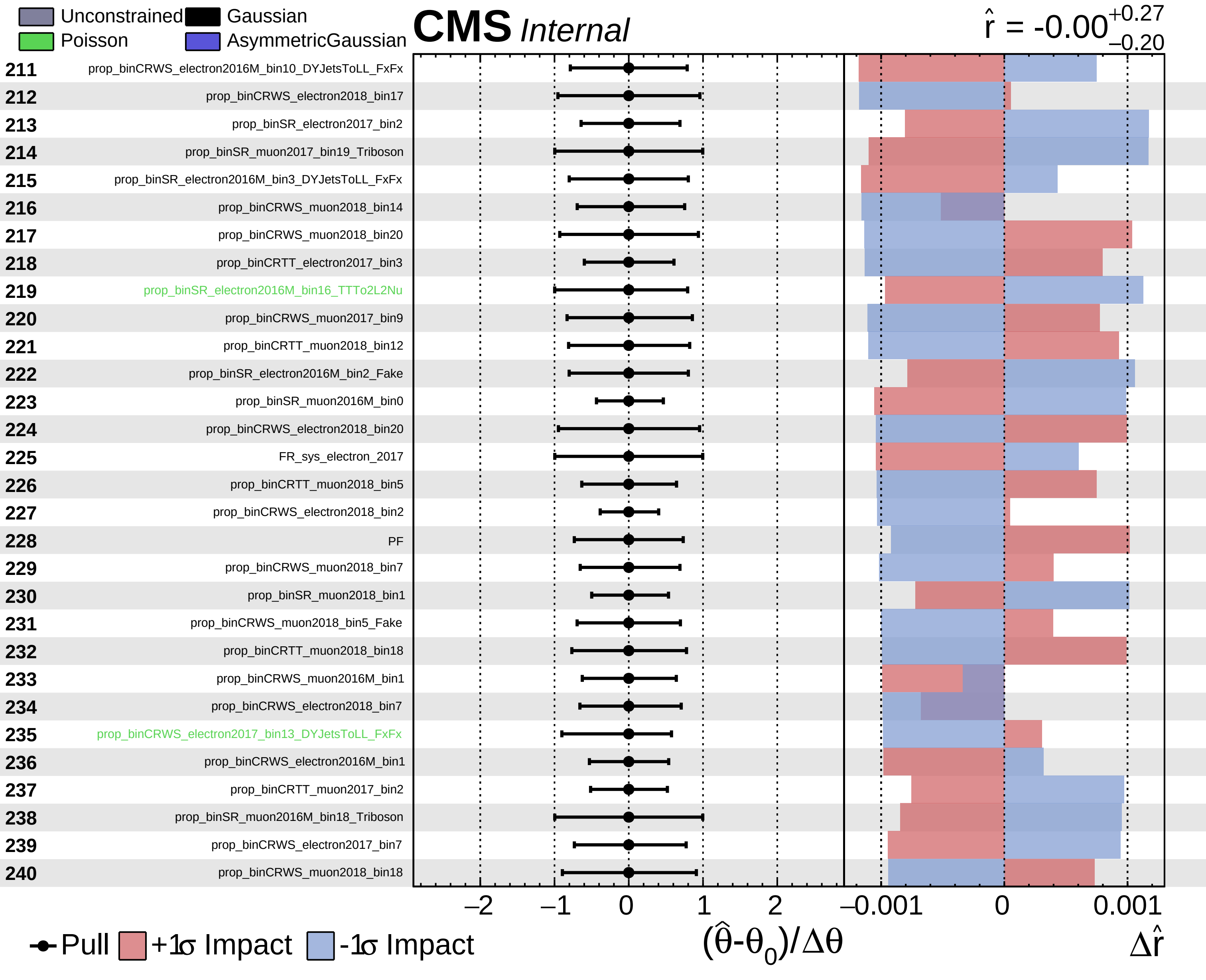
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
91 prop_binCRWS_electron2016M_bin3
92 prop_binSR_muon2016M_bin16
93 prop_binSR_muon2016M_bin15
94 prop_binSR_muon2016M_bin4
95 prop_binSR_electron2016M_bin18
96 prop_binCRWS_muon2018_bin3
97 prop_binSR_electron2018_bin14
98 prop_binSR_muon2016M_bin3
99 prop_binSR_electron2018_bin7
100 prop_binSR_electron2017_bin8
101 prop_binSR_electron2016M_bin10
102 prop_binSR_muon2017_bin3
103 prop_binSR_muon2016M_bin8
104 btag
105 TTest_muon_2018
$1.00^{+0.04}_{-0.04}$
106 prop_binSR_electron2018_bin12
107 prop_binCRTT_electron2017_bin0
108 prop_binSR_electron2017_bin16
109 prop_binSR_muon2017_bin19_TVX
110 prop_binSR_electron2018_bin6
111 prop_binCRWS_electron2017_bin1
112 prop_binSR_electron2017_bin12_ZZtoLep
113 prop_binCRTT_electron2018_bin2
114 prop_binSR_electron2017_bin7
115 prop_binCRWS_electron2017_bin10
116 prop_binSR_electron2017_bin15
117 prop_binCRWS_muon2018_bin8
118 prop_binCRWS_muon2016M_bin2
119 prop_binSR_electron2016M_bin17
120 prop_binSR_muon2016M_bin7
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

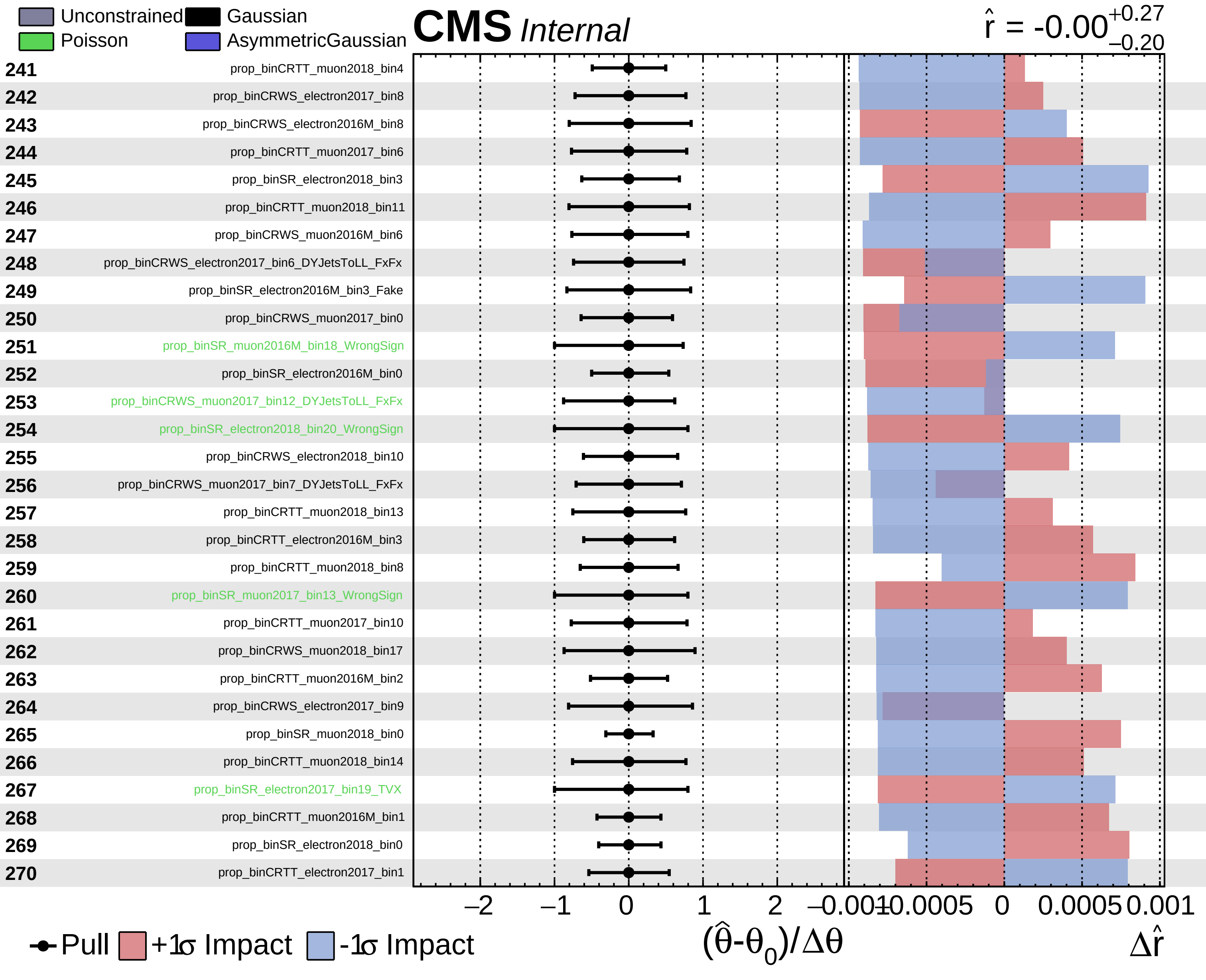

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.27 −0.20
121 prop_binSR_muon2017_bin11
122 prop_binSR_electron2017_bin11_Fake
123 prop_binSR_electron2018_bin4
124 prop_binCRWS_muon2018_bin13_DYJetsToLL_FxFx
125 prop_binSR_muon2016M_bin14
126 prop_binSR_muon2016M_bin13
127 prop_binCRWS_electron2018_bin5
128 prop_binCRWS_muon2017_bin8
129 prop_binCRWS_electron2018_bin9
130 prop_binSR_muon2016M_bin6
131 prop_binCRWS_electron2016M_bin6
132 prop_binSR_muon2016M_bin2
133 FR_sys_muon_2017
134 prop_binSR_electron2017_bin6
135 lumi_2016M
136 prop_binCRWS_electron2018_bin12_DYJetsToLL_FxFx
137 prop_binSR_electron2018_bin11
138 prop_binSR_electron2017_bin13
139 prop_binSR_muon2018_bin2
140 prop_binSR_muon2017_bin2
141 prop_binSR_electron2017_bin5
142 prop_binSR_electron2016M_bin9
143 prop_binCRWS_muon2017_bin4
144 prop_binSR_muon2018_bin3
145 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
146 prop_binCRWS_electron2018_bin0
147 prop_binCRWS_electron2018_bin13
148 prop_binCRWS_muon2017_bin13_DYJetsToLL_FxFx
149 prop_binCRWS_muon2018_bin0
150 metUnclust
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.27 −0.20
151 prop_binSR_muon2016M_bin12
152 prop_binSR_electron2018_bin20_Triboson
153 prop_binSR_muon2017_bin7
154 prop_binSR_electron2017_bin11_Triboson
155 prop_binSR_electron2016M_bin8
156 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
157 prop_binCRWS_muon2018_bin4
158 prop_binSR_electron2018_bin20_TVX
159 prop_binSR_muon2016M_bin5
160 prop_binSR_muon2016M_bin11
161 prop_binSR_muon2017_bin5
162 prop_binSR_muon2016M_bin18_TVX
163 prop_binSR_electron2016M_bin13
164 prop_binSR_electron2016M_bin15
165 FR_sys_electron_2018
166 prop_binCRTT_muon2018_bin0
167 prop_binSR_electron2016M_bin6
168 prop_binCRWS_electron2017_bin3
169 prop_binCRWS_muon2018_bin1
170 prop_binCRWS_muon2017_bin1
171 prop_binCRWS_muon2016M_bin7
172 FR_sys_muon_2016M
173 prop_binCRTT_electron2018_bin9
174 prop_binSR_electron2017_bin3
175 prop_binCRWS_muon2017_bin10
176 prop_binCRWS_electron2016M_bin2
177 prop_binCRTT_muon2018_bin9
178 prop_binCRWS_electron2018_bin11
179 prop_binCRTT_muon2018_bin2
180 prop_binCRWS_muon2018_bin6
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

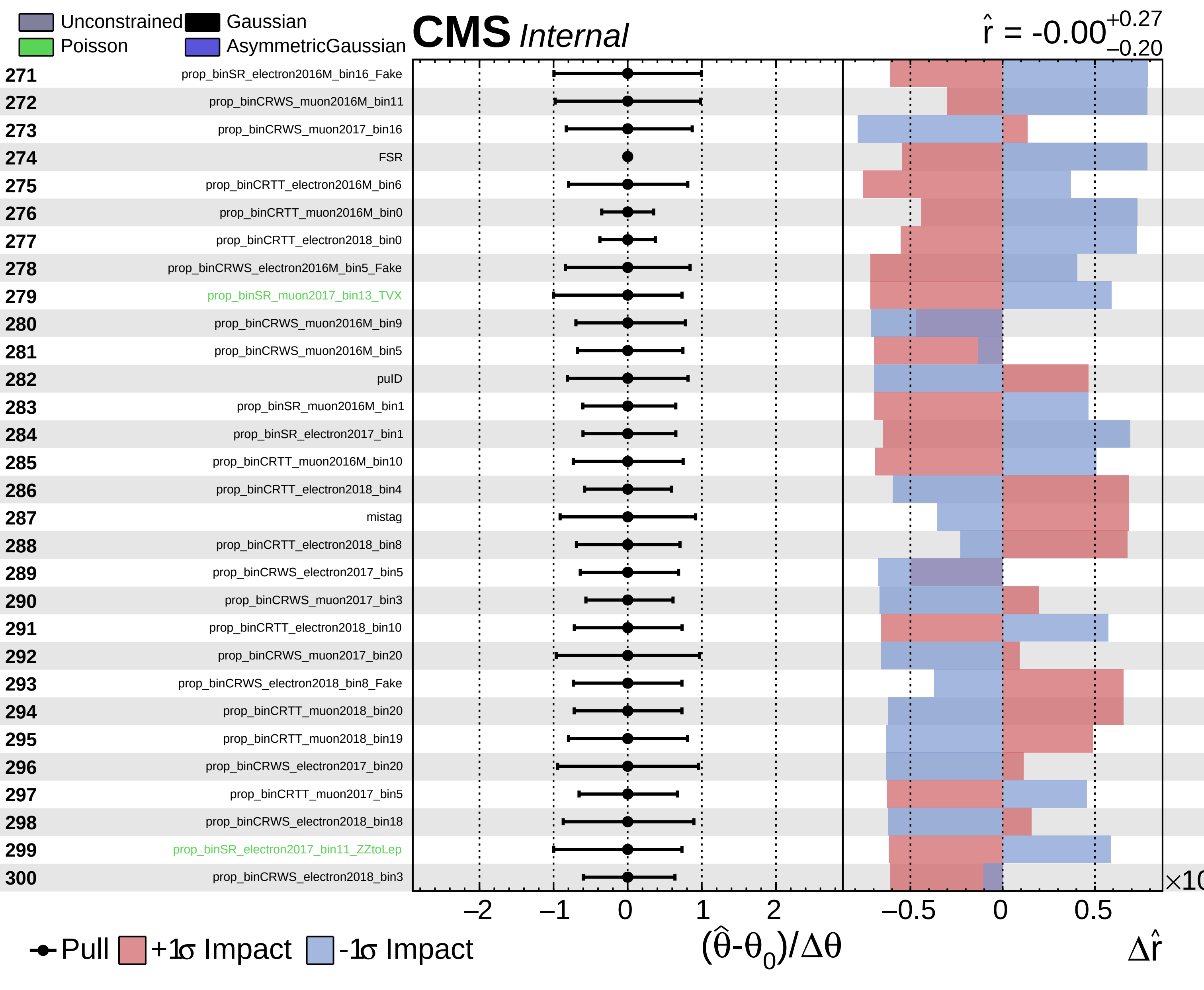


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
181 prop_binSR_electron2016M_bin7
182 prop_binCRWS_muon2016M_bin4
183 prop_binCRWS_electron2017_bin0
184 prop_binSR_electron2016M_bin11
185 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
186 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
187 prop_binSR_electron2016M_bin14
188 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
189 prop_binSR_muon2017_bin13_ZZtoLep
190 prop_binCRWS_muon2017_bin11_DYJetsToLL_FxFx
191 prop_binCRWS_electron2018_bin16_DYJetsToLL_FxFx
192 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
193 prop_binSR_electron2016M_bin12
194 prop_binCRWS_electron2017_bin2
195 prop_binCRWS_muon2016M_bin0
196 prop_binCRWS_muon2017_bin5
197 prop_binCRWS_muon2017_bin6
198 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
199 prop_binSR_electron2016M_bin5
200 prop_binCRTT_muon2017_bin0
201 prop_binCRWS_electron2018_bin1
202 prop_binCRWS_muon2018_bin9
203 prop_binSR_electron2016M_bin4
204 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
205 prop_binCRTT_muon2018_bin17
206 prop_binSR_muon2017_bin6_Fake
207 prop_binSR_electron2016M_bin1
208 prop_binSR_electron2018_bin2
209 prop_binCRWS_muon2018_bin15_DYJetsToLL_FxFx
210 prop_binSR_muon2017_bin1
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

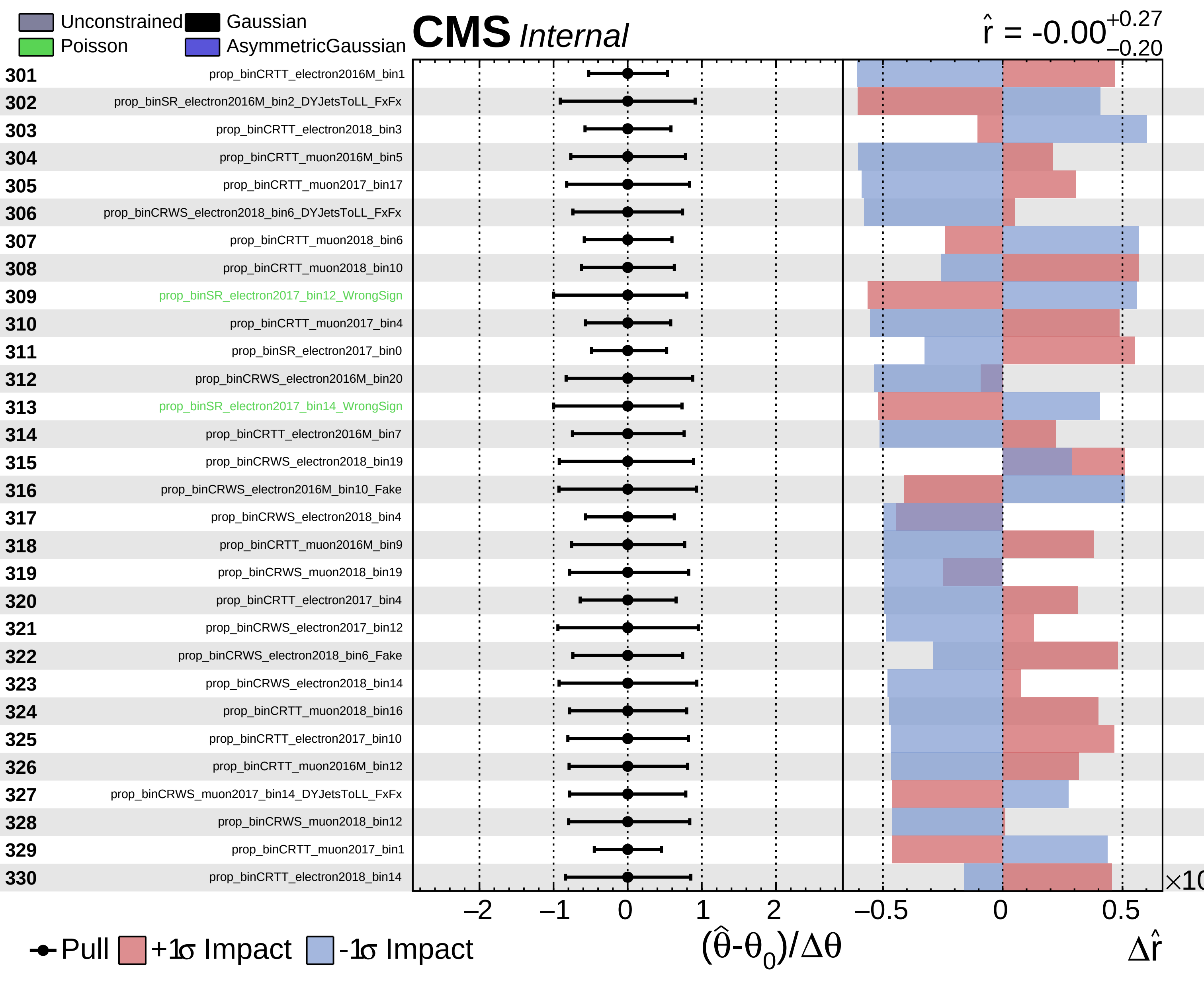

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
211 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
212 prop_binCRWS_electron2018_bin17
213 prop_binSR_electron2017_bin2
214 prop_binSR_muon2017_bin19_Triboson
215 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
216 prop_binCRWS_muon2018_bin14
217 prop_binCRWS_muon2018_bin20
218 prop_binCRTT_electron2017_bin3
219 prop_binSR_electron2016M_bin16_TTTo2L2Nu
220 prop_binCRWS_muon2017_bin9
221 prop_binCRTT_muon2018_bin12
222 prop_binSR_electron2016M_bin2_Fake
223 prop_binSR_muon2016M_bin0
224 prop_binCRWS_electron2018_bin20
225 FR_sys_electron_2017
226 prop_binCRTT_muon2018_bin5
227 prop_binCRWS_electron2018_bin2
228 PF
229 prop_binCRWS_muon2018_bin7
230 prop_binSR_muon2018_bin1
231 prop_binCRWS_muon2018_bin5_Fake
232 prop_binCRTT_muon2018_bin18
233 prop_binCRWS_muon2016M_bin1
234 prop_binCRWS_electron2018_bin7
235 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
236 prop_binCRWS_electron2016M_bin1
237 prop_binCRTT_muon2017_bin2
238 prop_binSR_muon2016M_bin18_Triboson
239 prop_binCRWS_electron2017_bin7
240 prop_binCRWS_muon2018_bin18
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



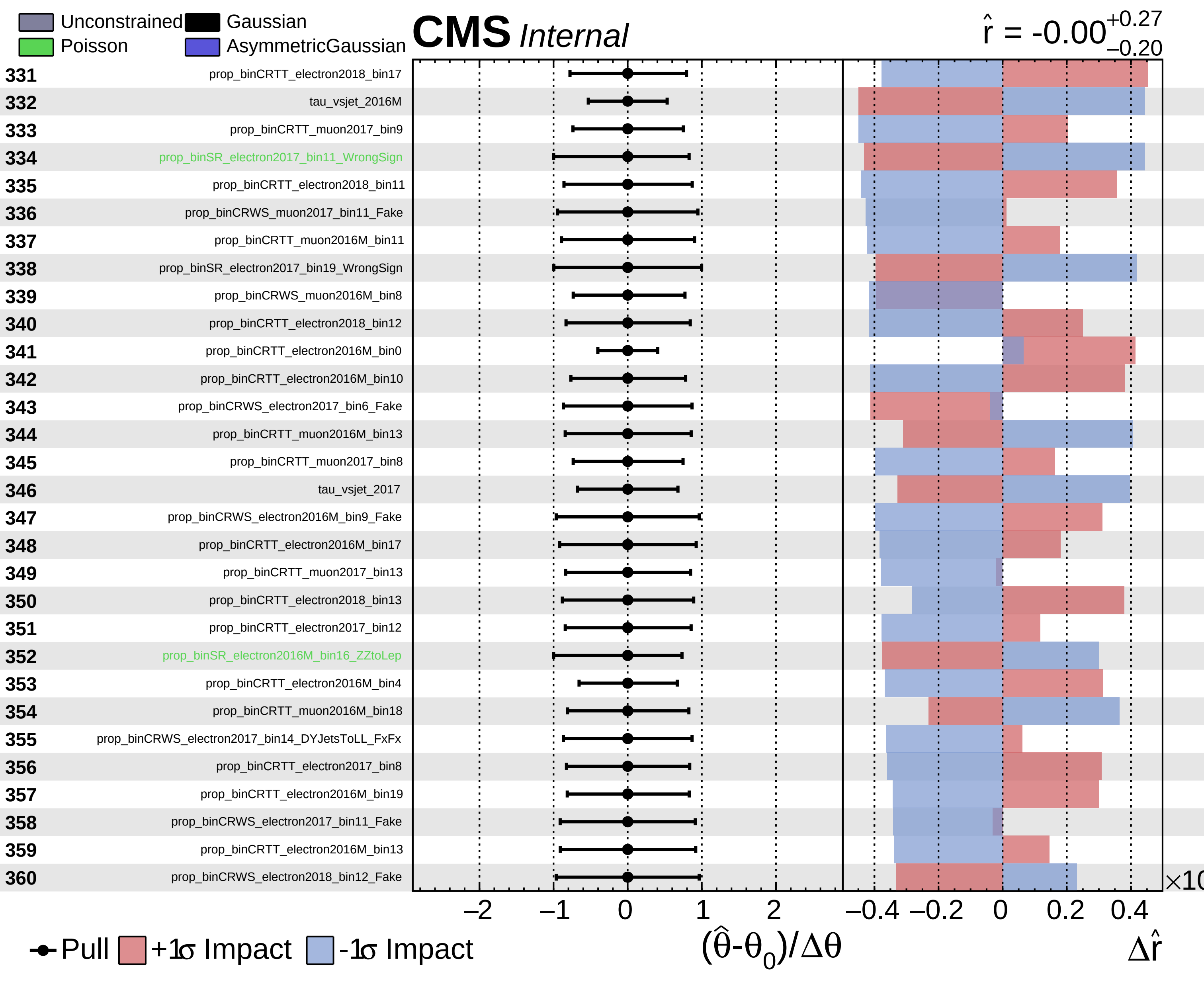
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
241 prop_binCRTT_muon2018_bin4
242 prop_binCRWS_electron2017_bin8
243 prop_binCRWS_electron2016M_bin8
244 prop_binCRTT_muon2017_bin6
245 prop_binSR_electron2018_bin3
246 prop_binCRTT_muon2018_bin11
247 prop_binCRWS_muon2016M_bin6
248 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
249 prop_binSR_electron2016M_bin3_Fake
250 prop_binCRWS_muon2017_bin0
251 prop_binSR_muon2016M_bin18_WrongSign
252 prop_binSR_electron2016M_bin0
253 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
254 prop_binSR_electron2018_bin20_WrongSign
255 prop_binCRWS_electron2018_bin10
256 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
257 prop_binCRTT_muon2018_bin13
258 prop_binCRTT_electron2016M_bin3
259 prop_binCRTT_muon2018_bin8
260 prop_binSR_muon2017_bin13_WrongSign
261 prop_binCRTT_muon2017_bin10
262 prop_binCRWS_muon2018_bin17
263 prop_binCRTT_muon2016M_bin2
264 prop_binCRWS_electron2017_bin9
265 prop_binSR_muon2018_bin0
266 prop_binCRTT_muon2018_bin14
267 prop_binSR_electron2017_bin19_TVX
268 prop_binCRTT_muon2016M_bin1
269 prop_binSR_electron2018_bin0
270 prop_binCRTT_electron2017_bin1
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



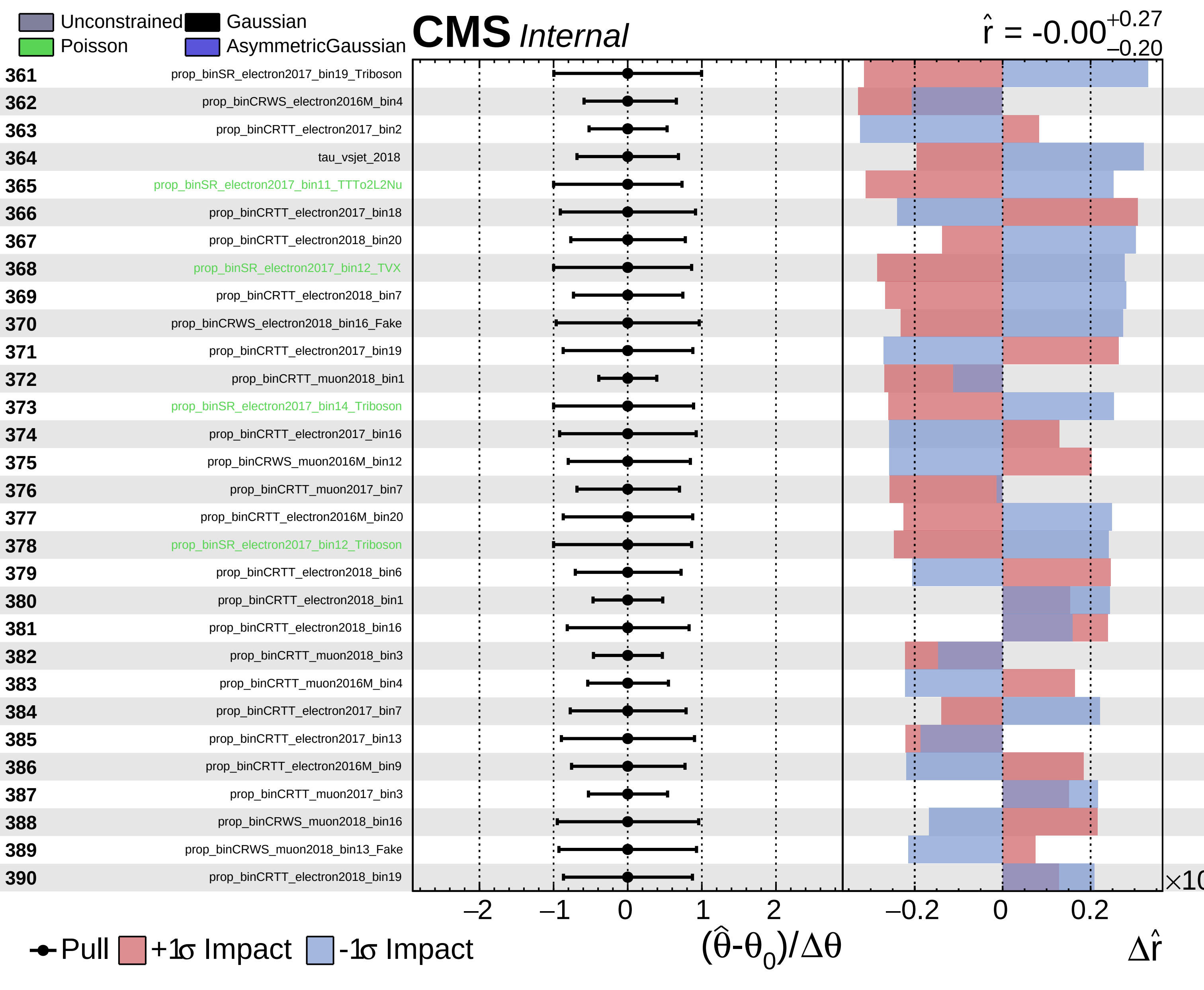
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
271 prop_binSR_electron2016M_bin16_Fake
272 prop_binCRWS_muon2016M_bin11
273 prop_binCRWS_muon2017_bin16
274 FSR
275 prop_binCRTT_electron2016M_bin6
276 prop_binCRTT_muon2016M_bin0
277 prop_binCRTT_electron2018_bin0
278 prop_binCRWS_electron2016M_bin5_Fake
279 prop_binSR_muon2017_bin13_TVX
280 prop_binCRWS_muon2016M_bin9
281 prop_binCRWS_muon2016M_bin5
282 puID
283 prop_binSR_muon2016M_bin1
284 prop_binSR_electron2017_bin1
285 prop_binCRTT_muon2016M_bin10
286 prop_binCRTT_electron2018_bin4
287 mistag
288 prop_binCRTT_electron2018_bin8
289 prop_binCRWS_electron2017_bin5
290 prop_binCRWS_muon2017_bin3
291 prop_binCRTT_electron2018_bin10
292 prop_binCRWS_muon2017_bin20
293 prop_binCRWS_electron2018_bin8_Fake
294 prop_binCRTT_muon2018_bin20
295 prop_binCRTT_muon2018_bin19
296 prop_binCRWS_electron2017_bin20
297 prop_binCRTT_muon2017_bin5
298 prop_binCRWS_electron2018_bin18
299 prop_binSR_electron2017_bin11_ZZtoLep
300 prop_binCRWS_electron2018_bin3
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

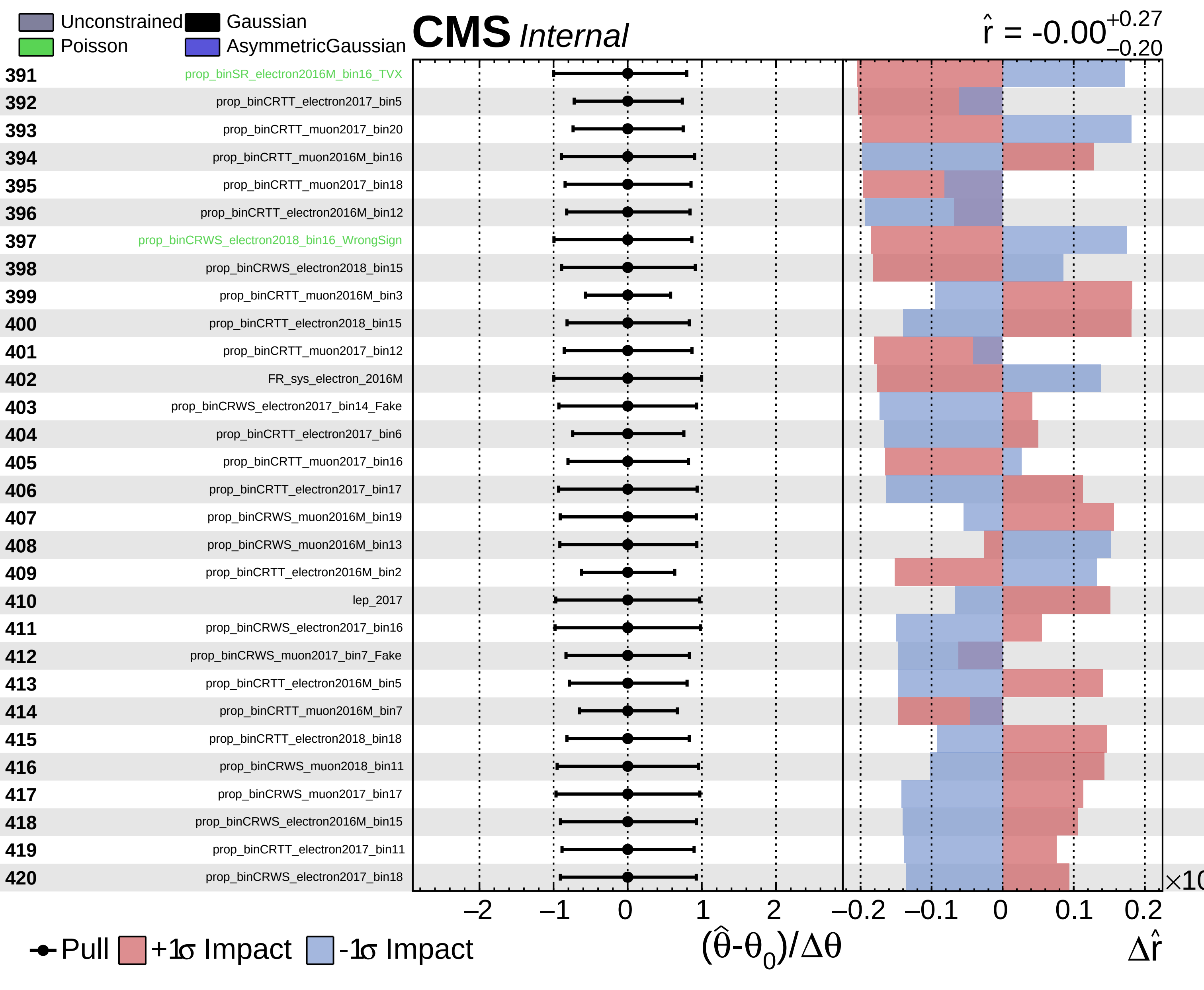


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
301 prop_binCRTT_electron2016M_bin1
302 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
303 prop_binCRTT_electron2018_bin3
304 prop_binCRTT_muon2016M_bin5
305 prop_binCRTT_muon2017_bin17
306 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
307 prop_binCRTT_muon2018_bin6
308 prop_binCRTT_muon2018_bin10
309 prop_binSR_electron2017_bin12_WrongSign
310 prop_binCRTT_muon2017_bin4
311 prop_binSR_electron2017_bin0
312 prop_binCRWS_electron2016M_bin20
313 prop_binSR_electron2017_bin14_WrongSign
314 prop_binCRTT_electron2016M_bin7
315 prop_binCRWS_electron2018_bin19
316 prop_binCRWS_electron2016M_bin10_Fake
317 prop_binCRWS_electron2018_bin4
318 prop_binCRTT_muon2016M_bin9
319 prop_binCRWS_muon2018_bin19
320 prop_binCRTT_electron2017_bin4
321 prop_binCRWS_electron2017_bin12
322 prop_binCRWS_electron2018_bin6_Fake
323 prop_binCRWS_electron2018_bin14
324 prop_binCRTT_muon2018_bin16
325 prop_binCRTT_electron2017_bin10
326 prop_binCRTT_muon2016M_bin12
327 prop_binCRWS_muon2017_bin14_DYJetsToLL_FxFx
328 prop_binCRWS_muon2018_bin12
329 prop_binCRTT_muon2017_bin1
330 prop_binCRTT_electron2018_bin14
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

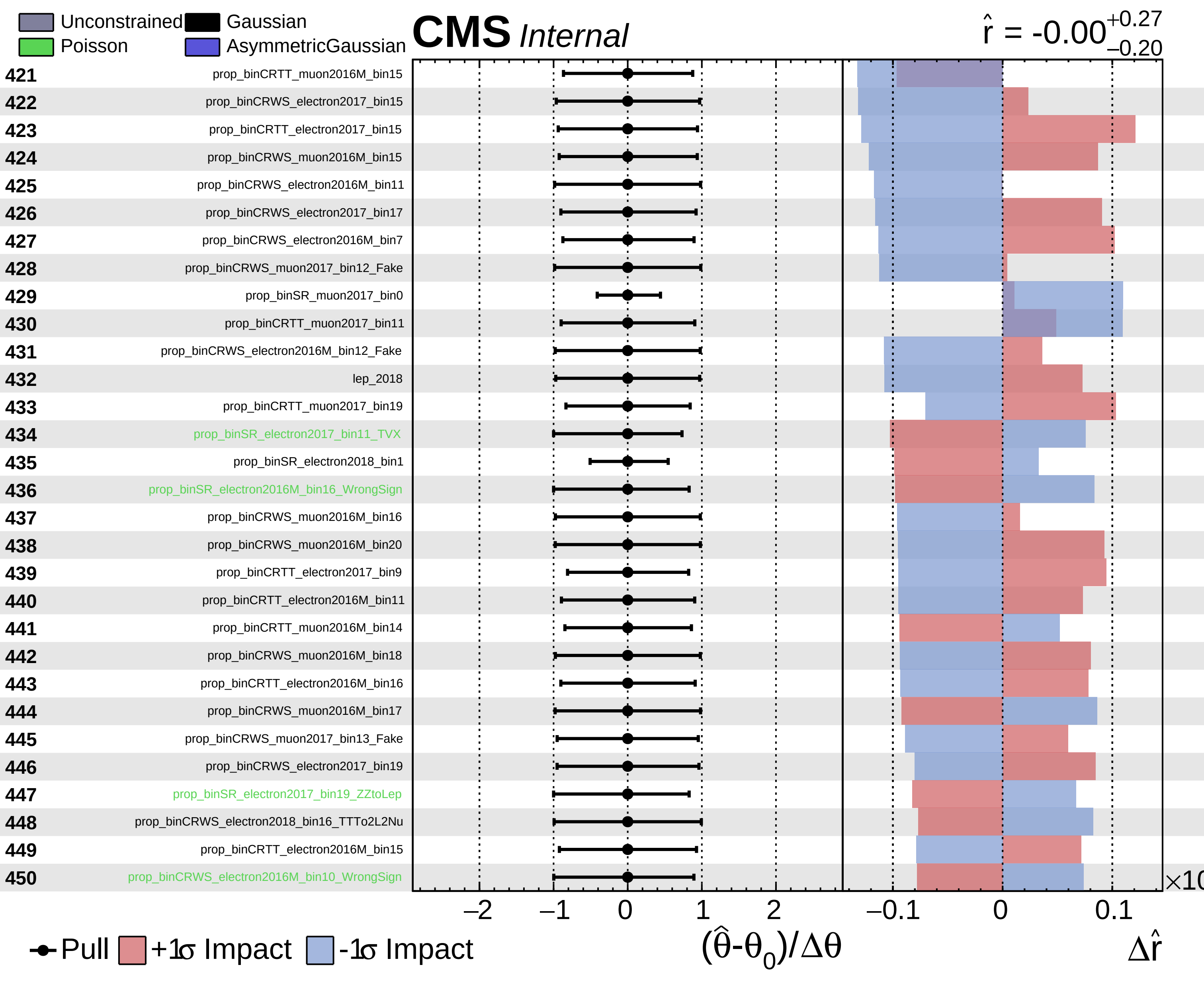


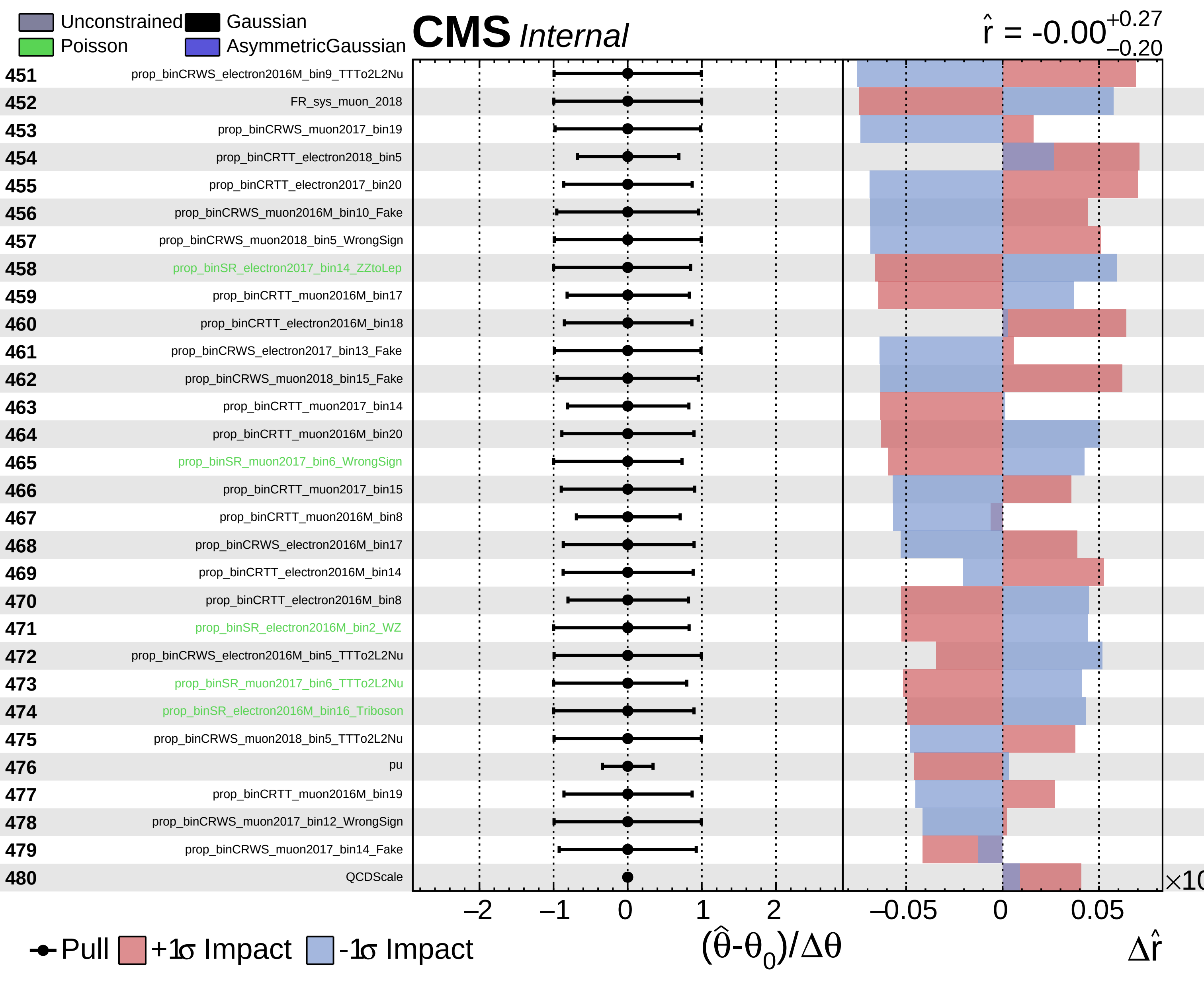
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.27 −0.20
331 prop_binCRTT_electron2018_bin17
332 tau_vsjet_2016M
333 prop_binCRTT_muon2017_bin9
334 prop_binSR_electron2017_bin11_WrongSign
335 prop_binCRTT_electron2018_bin11
336 prop_binCRWS_muon2017_bin11_Fake
337 prop_binCRTT_muon2016M_bin11
338 prop_binSR_electron2017_bin19_WrongSign
339 prop_binCRWS_muon2016M_bin8
340 prop_binCRTT_electron2018_bin12
341 prop_binCRTT_electron2016M_bin0
342 prop_binCRTT_electron2016M_bin10
343 prop_binCRWS_electron2017_bin6_Fake
344 prop_binCRTT_muon2016M_bin13
345 prop_binCRTT_muon2017_bin8
346 tau_vsjet_2017
347 prop_binCRWS_electron2016M_bin9_Fake
348 prop_binCRTT_electron2016M_bin17
349 prop_binCRTT_muon2017_bin13
350 prop_binCRTT_electron2018_bin13
351 prop_binCRTT_electron2017_bin12
352 prop_binSR_electron2016M_bin16_ZZtoLep
353 prop_binCRTT_electron2016M_bin4
354 prop_binCRTT_muon2016M_bin18
355 prop_binCRWS_electron2017_bin14_DYJetsToLL_FxFx
356 prop_binCRTT_electron2017_bin8
357 prop_binCRTT_electron2016M_bin19
358 prop_binCRWS_electron2017_bin11_Fake
359 prop_binCRTT_electron2016M_bin13
360 prop_binCRWS_electron2018_bin12_Fake
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
361 prop_binSR_electron2017_bin19_Triboson
362 prop_binCRWS_electron2016M_bin4
363 prop_binCRTT_electron2017_bin2
364 tau_vsjet_2018
365 prop_binSR_electron2017_bin11_TTTo2L2Nu
366 prop_binCRTT_electron2017_bin18
367 prop_binCRTT_electron2018_bin20
368 prop_binSR_electron2017_bin12_TVX
369 prop_binCRTT_electron2018_bin7
370 prop_binCRWS_electron2018_bin16_Fake
371 prop_binCRTT_electron2017_bin19
372 prop_binCRTT_muon2018_bin1
373 prop_binSR_electron2017_bin14_Triboson
374 prop_binCRTT_electron2017_bin16
375 prop_binCRWS_muon2016M_bin12
376 prop_binCRTT_muon2017_bin7
377 prop_binCRTT_electron2016M_bin20
378 prop_binSR_electron2017_bin12_Triboson
379 prop_binCRTT_electron2018_bin6
380 prop_binCRTT_electron2018_bin1
381 prop_binCRTT_electron2018_bin16
382 prop_binCRTT_muon2018_bin3
383 prop_binCRTT_muon2016M_bin4
384 prop_binCRTT_electron2017_bin7
385 prop_binCRTT_electron2017_bin13
386 prop_binCRTT_electron2016M_bin9
387 prop_binCRTT_muon2017_bin3
388 prop_binCRWS_muon2018_bin16
389 prop_binCRWS_muon2018_bin13_Fake
390 prop_binCRTT_electron2018_bin19
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

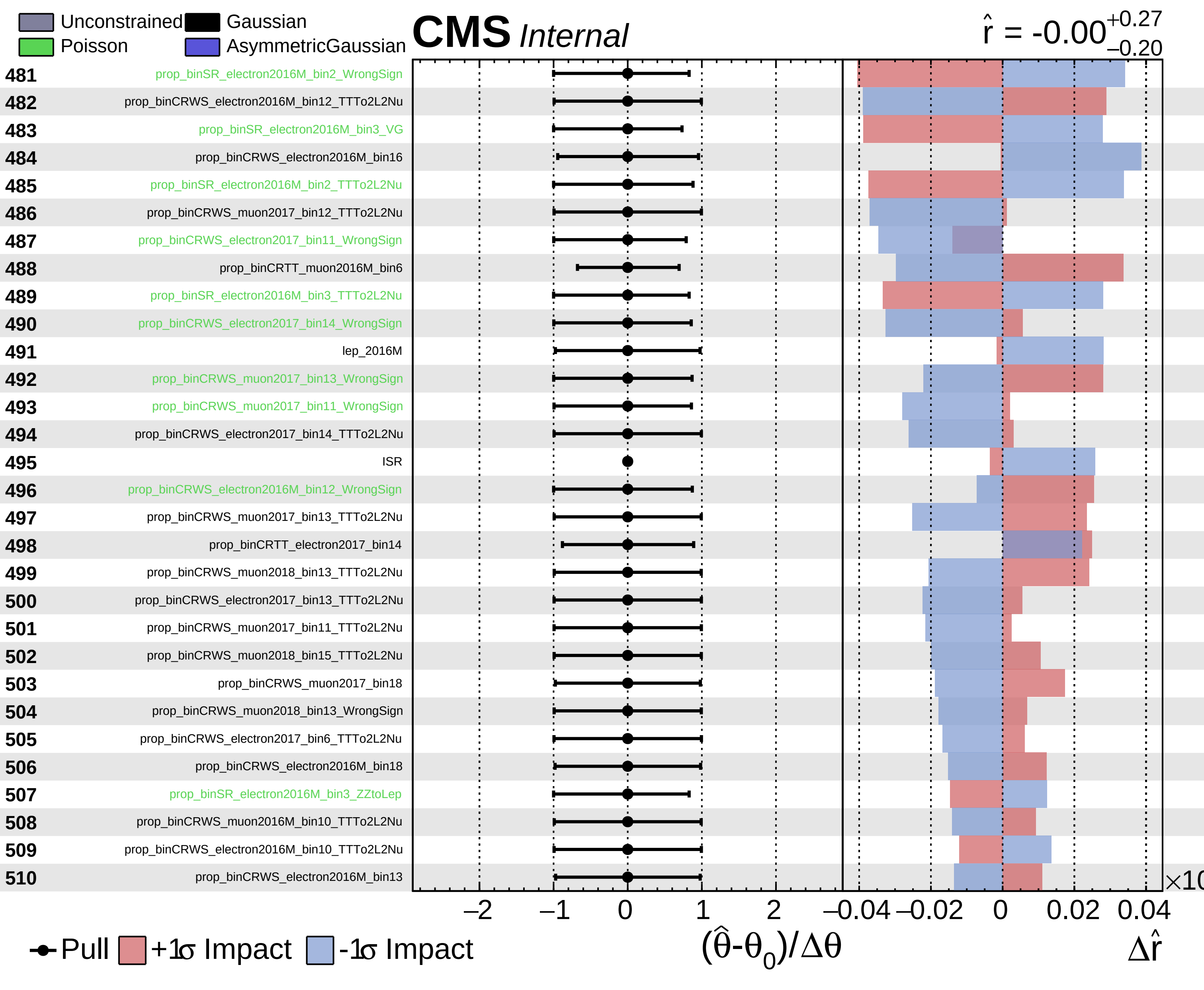


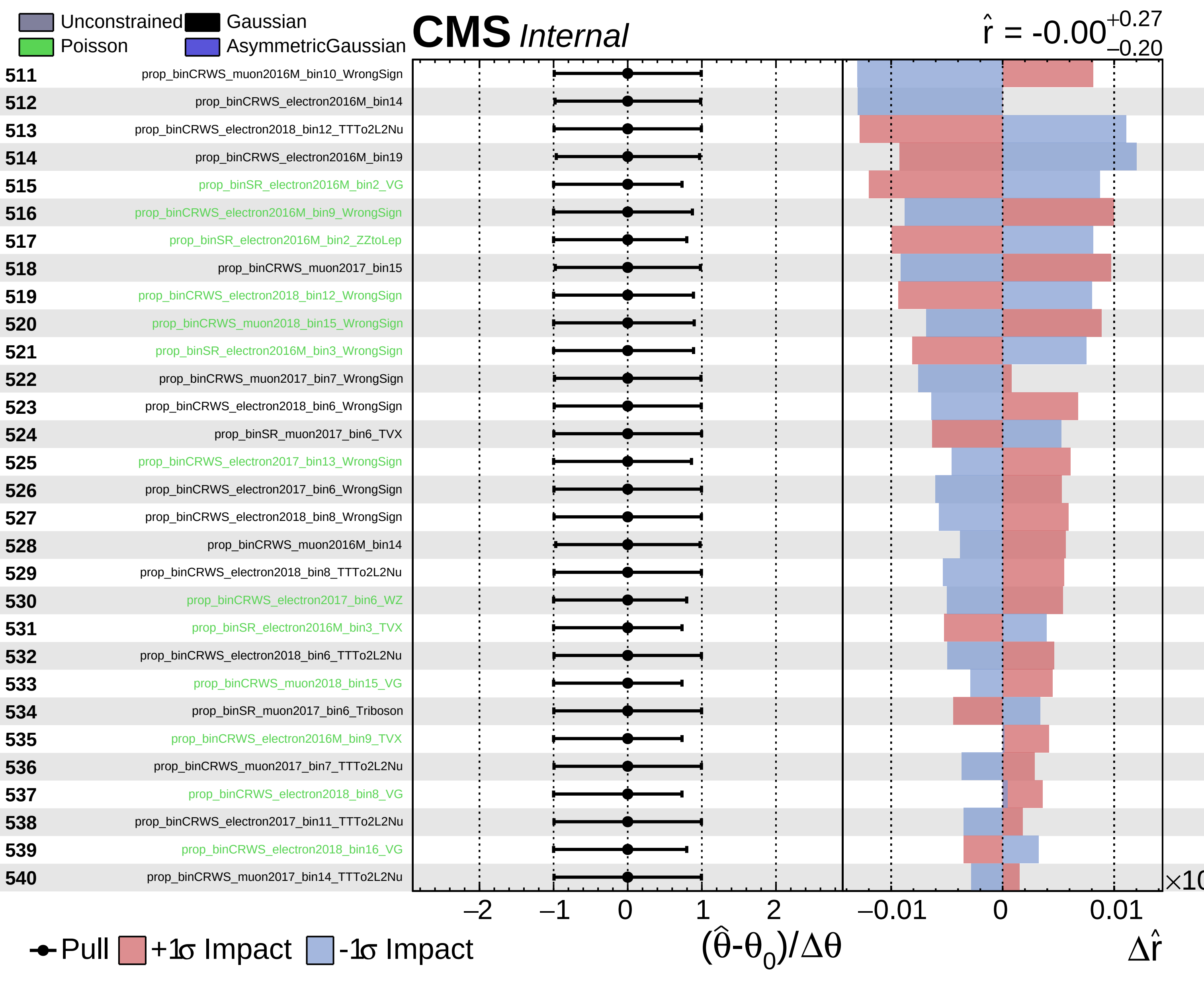
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
391 prop_binSR_electron2016M_bin16_TVX
392 prop_binCRTT_electron2017_bin5
393 prop_binCRTT_muon2017_bin20
394 prop_binCRTT_muon2016M_bin16
395 prop_binCRTT_muon2017_bin18
396 prop_binCRTT_electron2016M_bin12
397 prop_binCRWS_electron2018_bin16_WrongSign
398 prop_binCRWS_electron2018_bin15
399 prop_binCRTT_muon2016M_bin3
400 prop_binCRTT_electron2018_bin15
401 prop_binCRTT_muon2017_bin12
402 FR_sys_electron_2016M
403 prop_binCRWS_electron2017_bin14_Fake
404 prop_binCRTT_electron2017_bin6
405 prop_binCRTT_muon2017_bin16
406 prop_binCRTT_electron2017_bin17
407 prop_binCRWS_muon2016M_bin19
408 prop_binCRWS_muon2016M_bin13
409 prop_binCRTT_electron2016M_bin2
410 lep_2017
411 prop_binCRWS_electron2017_bin16
412 prop_binCRWS_muon2017_bin7_Fake
413 prop_binCRTT_electron2016M_bin5
414 prop_binCRTT_muon2016M_bin7
415 prop_binCRTT_electron2018_bin18
416 prop_binCRWS_muon2018_bin11
417 prop_binCRWS_muon2017_bin17
418 prop_binCRWS_electron2016M_bin15
419 prop_binCRTT_electron2017_bin11
420 prop_binCRWS_electron2017_bin18
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
421 prop_binCRTT_muon2016M_bin15
422 prop_binCRWS_electron2017_bin15
423 prop_binCRTT_electron2017_bin15
424 prop_binCRWS_muon2016M_bin15
425 prop_binCRWS_electron2016M_bin11
426 prop_binCRWS_electron2017_bin17
427 prop_binCRWS_electron2016M_bin7
428 prop_binCRWS_muon2017_bin12_Fake
429 prop_binSR_muon2017_bin0
430 prop_binCRTT_muon2017_bin11
431 prop_binCRWS_electron2016M_bin12_Fake
432 lep_2018
433 prop_binCRTT_muon2017_bin19
434 prop_binSR_electron2017_bin11_TVX
435 prop_binSR_electron2018_bin1
436 prop_binSR_electron2016M_bin16_WrongSign
437 prop_binCRWS_muon2016M_bin16
438 prop_binCRWS_muon2016M_bin20
439 prop_binCRTT_electron2017_bin9
440 prop_binCRTT_electron2016M_bin11
441 prop_binCRTT_muon2016M_bin14
442 prop_binCRWS_muon2016M_bin18
443 prop_binCRTT_electron2016M_bin16
444 prop_binCRWS_muon2016M_bin17
445 prop_binCRWS_muon2017_bin13_Fake
446 prop_binCRWS_electron2017_bin19
447 prop_binSR_electron2017_bin19_ZZtoLep
448 prop_binCRWS_electron2018_bin16_TTTo2L2Nu
449 prop_binCRTT_electron2016M_bin15
450 prop_binCRWS_electron2016M_bin10_WrongSign
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

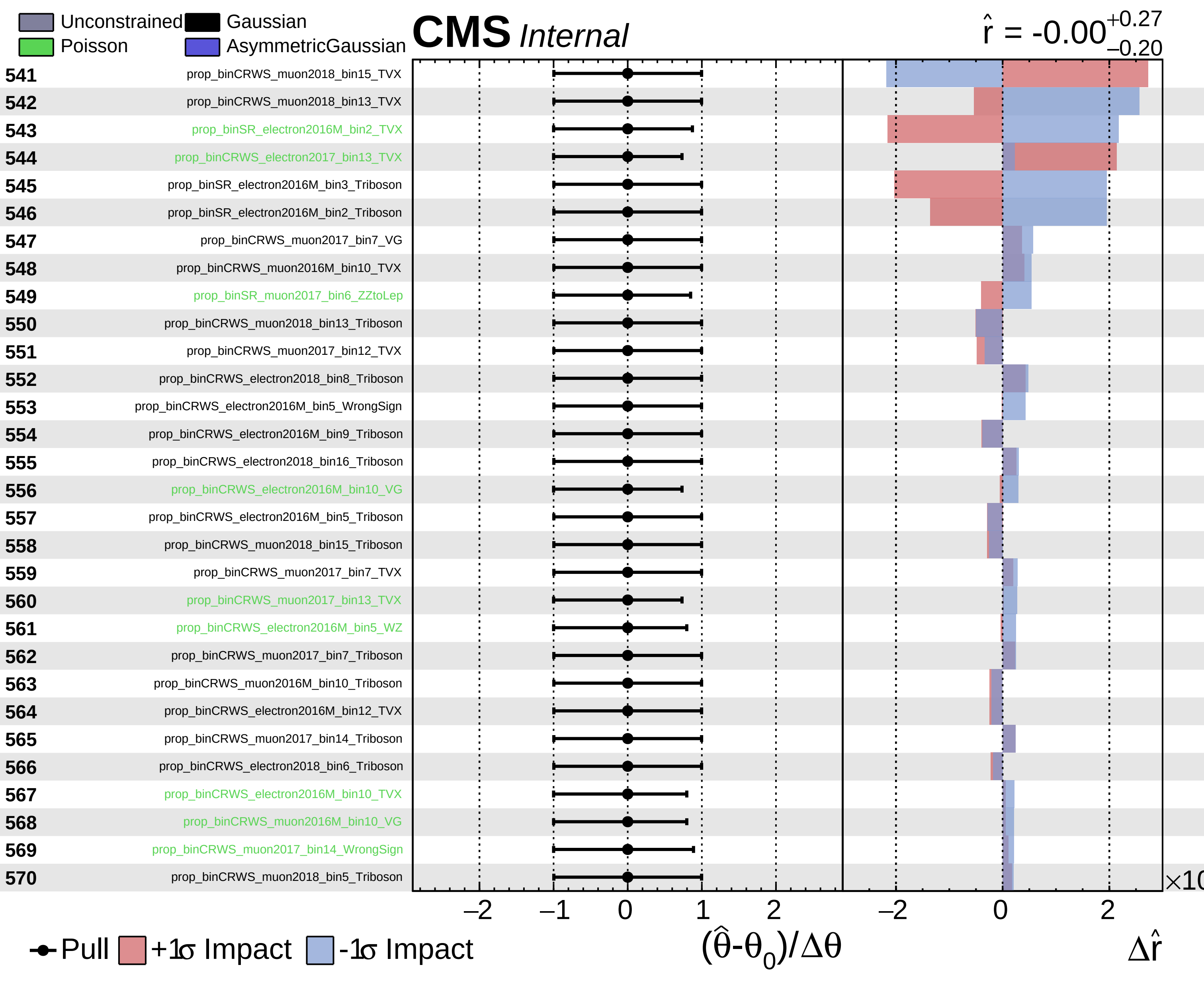


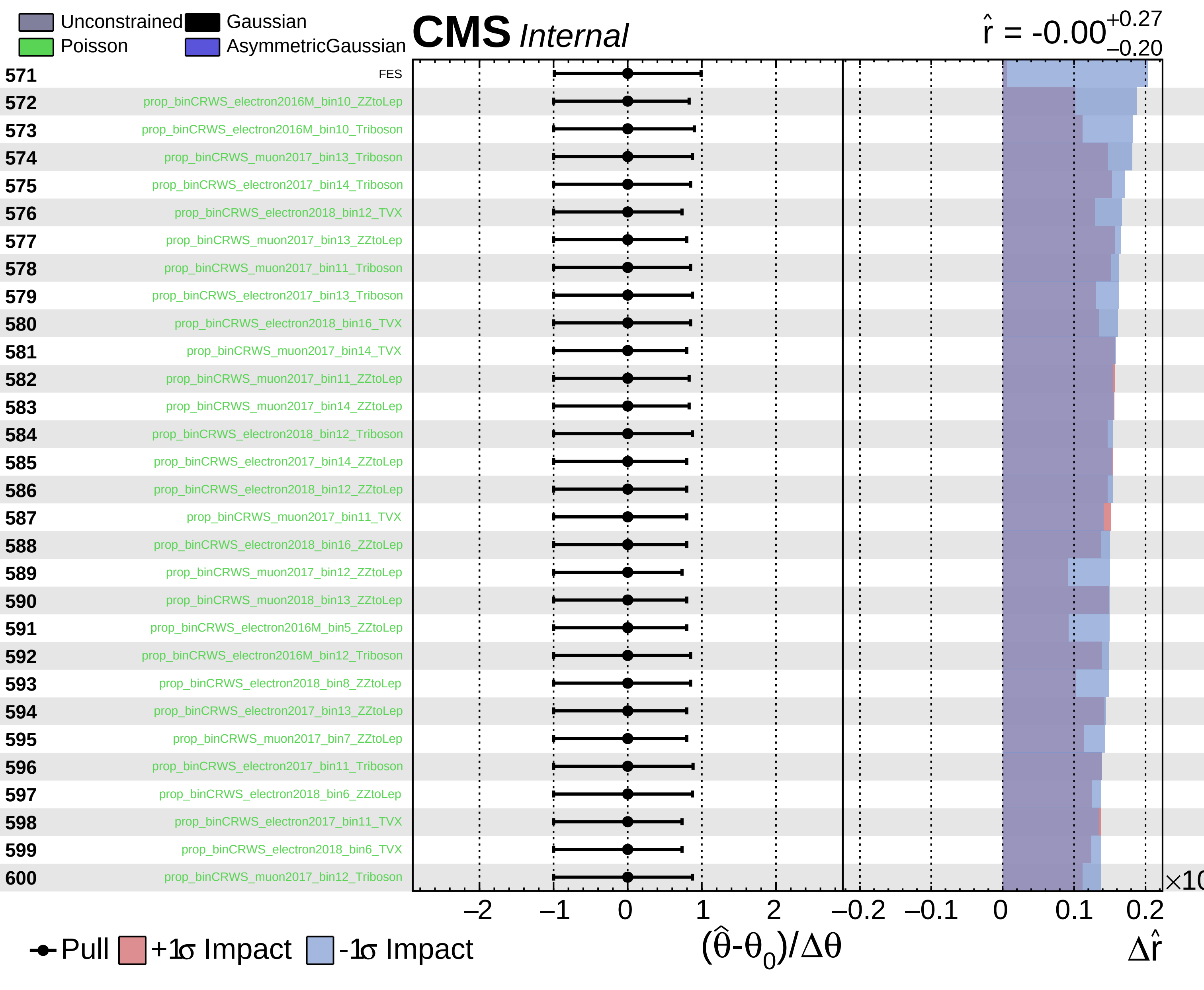
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
451 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
452 FR_sys_muon_2018
453 prop_binCRWS_muon2017_bin19
454 prop_binCRTT_electron2018_bin5
455 prop_binCRTT_electron2017_bin20
456 prop_binCRWS_muon2016M_bin10_Fake
457 prop_binCRWS_muon2018_bin5_WrongSign
458 prop_binSR_electron2017_bin14_ZZtoLep
459 prop_binCRTT_muon2016M_bin17
460 prop_binCRTT_electron2016M_bin18
461 prop_binCRWS_electron2017_bin13_Fake
462 prop_binCRWS_muon2018_bin15_Fake
463 prop_binCRTT_muon2017_bin14
464 prop_binCRTT_muon2016M_bin20
465 prop_binSR_muon2017_bin6_WrongSign
466 prop_binCRTT_muon2017_bin15
467 prop_binCRTT_muon2016M_bin8
468 prop_binCRWS_electron2016M_bin17
469 prop_binCRTT_electron2016M_bin14
470 prop_binCRTT_electron2016M_bin8
471 prop_binSR_electron2016M_bin2_WZ
472 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
473 prop_binSR_muon2017_bin6_TTTo2L2Nu
474 prop_binSR_electron2016M_bin16_Triboson
475 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
476 pu
477 prop_binCRTT_muon2016M_bin19
478 prop_binCRWS_muon2017_bin12_WrongSign
479 prop_binCRWS_muon2017_bin14_Fake
480 QCDScale
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
481 prop_binSR_electron2016M_bin2_WrongSign
482 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
483 prop_binSR_electron2016M_bin3_VG
484 prop_binCRWS_electron2016M_bin16
485 prop_binSR_electron2016M_bin2_TTTo2L2Nu
486 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
487 prop_binCRWS_electron2017_bin11_WrongSign
488 prop_binCRTT_muon2016M_bin6
489 prop_binSR_electron2016M_bin3_TTTo2L2Nu
490 prop_binCRWS_electron2017_bin14_WrongSign
491 lep_2016M
492 prop_binCRWS_muon2017_bin13_WrongSign
493 prop_binCRWS_muon2017_bin11_WrongSign
494 prop_binCRWS_electron2017_bin14_TTTo2L2Nu
495 ISR
496 prop_binCRWS_electron2016M_bin12_WrongSign
497 prop_binCRWS_muon2017_bin13_TTTo2L2Nu
498 prop_binCRTT_electron2017_bin14
499 prop_binCRWS_muon2018_bin13_TTTo2L2Nu
500 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
501 prop_binCRWS_muon2017_bin11_TTTo2L2Nu
502 prop_binCRWS_muon2018_bin15_TTTo2L2Nu
503 prop_binCRWS_muon2017_bin18
504 prop_binCRWS_muon2018_bin13_WrongSign
505 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
506 prop_binCRWS_electron2016M_bin18
507 prop_binSR_electron2016M_bin3_ZZtoLep
508 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
509 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
510 prop_binCRWS_electron2016M_bin13
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

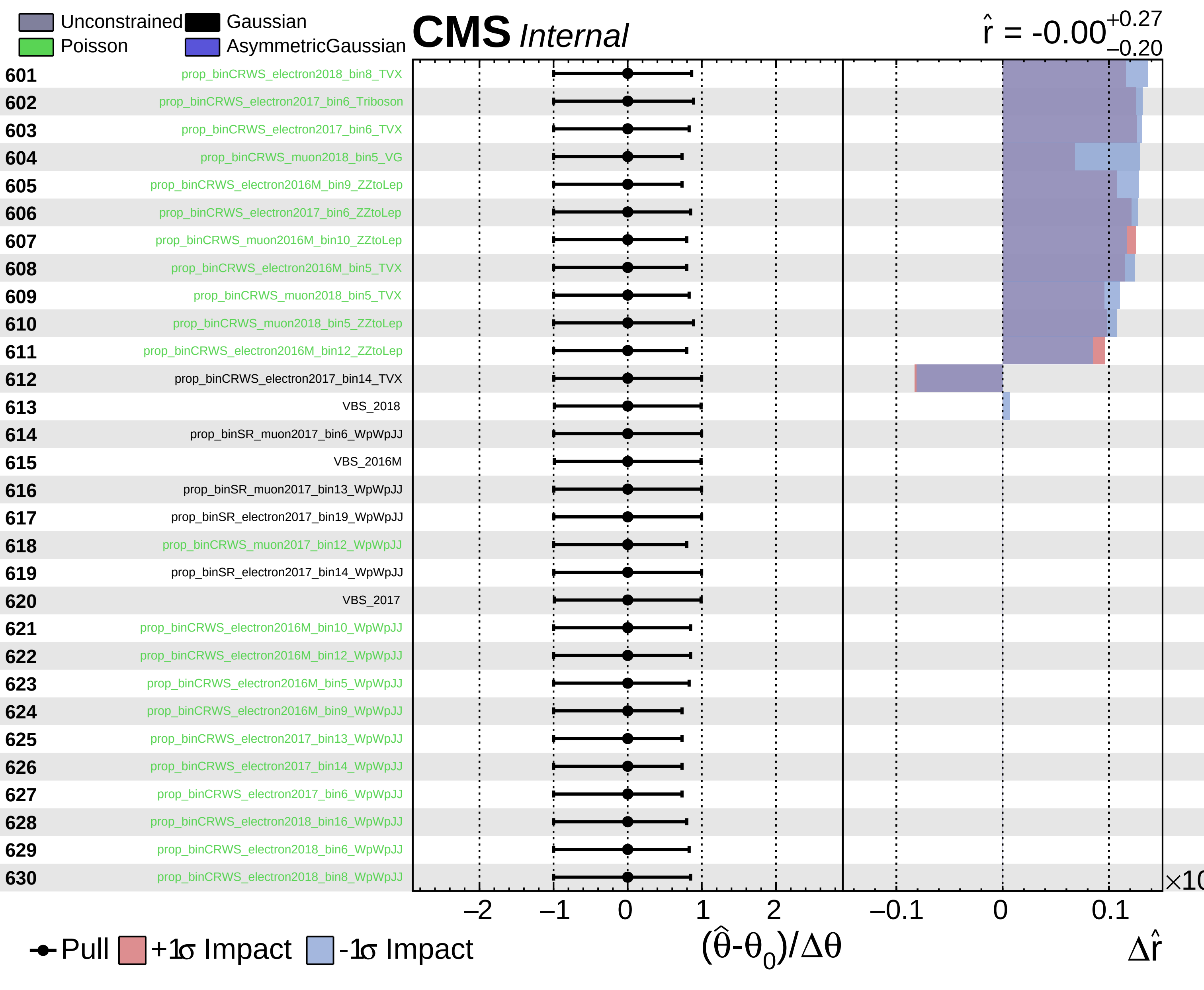


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
511 prop_binCRWS_muon2016M_bin10_WrongSign
512 prop_binCRWS_electron2016M_bin14
513 prop_binCRWS_electron2018_bin12_TTTo2L2Nu
514 prop_binCRWS_electron2016M_bin19
515 prop_binSR_electron2016M_bin2_VG
516 prop_binCRWS_electron2016M_bin9_WrongSign
517 prop_binSR_electron2016M_bin2_ZZtoLep
518 prop_binCRWS_muon2017_bin15
519 prop_binCRWS_electron2018_bin12_WrongSign
520 prop_binCRWS_muon2018_bin15_WrongSign
521 prop_binSR_electron2016M_bin3_WrongSign
522 prop_binCRWS_muon2017_bin7_WrongSign
523 prop_binCRWS_electron2018_bin6_WrongSign
524 prop_binSR_muon2017_bin6_TVX
525 prop_binCRWS_electron2017_bin13_WrongSign
526 prop_binCRWS_electron2017_bin6_WrongSign
527 prop_binCRWS_electron2018_bin8_WrongSign
528 prop_binCRWS_muon2016M_bin14
529 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
530 prop_binCRWS_electron2017_bin6_WZ
531 prop_binSR_electron2016M_bin3_TVX
532 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
533 prop_binCRWS_muon2018_bin15_VG
534 prop_binSR_muon2017_bin6_Triboson
535 prop_binCRWS_electron2016M_bin9_TVX
536 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
537 prop_binCRWS_electron2018_bin8_VG
538 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
539 prop_binCRWS_electron2018_bin16_VG
540 prop_binCRWS_muon2017_bin14_TTTo2L2Nu
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

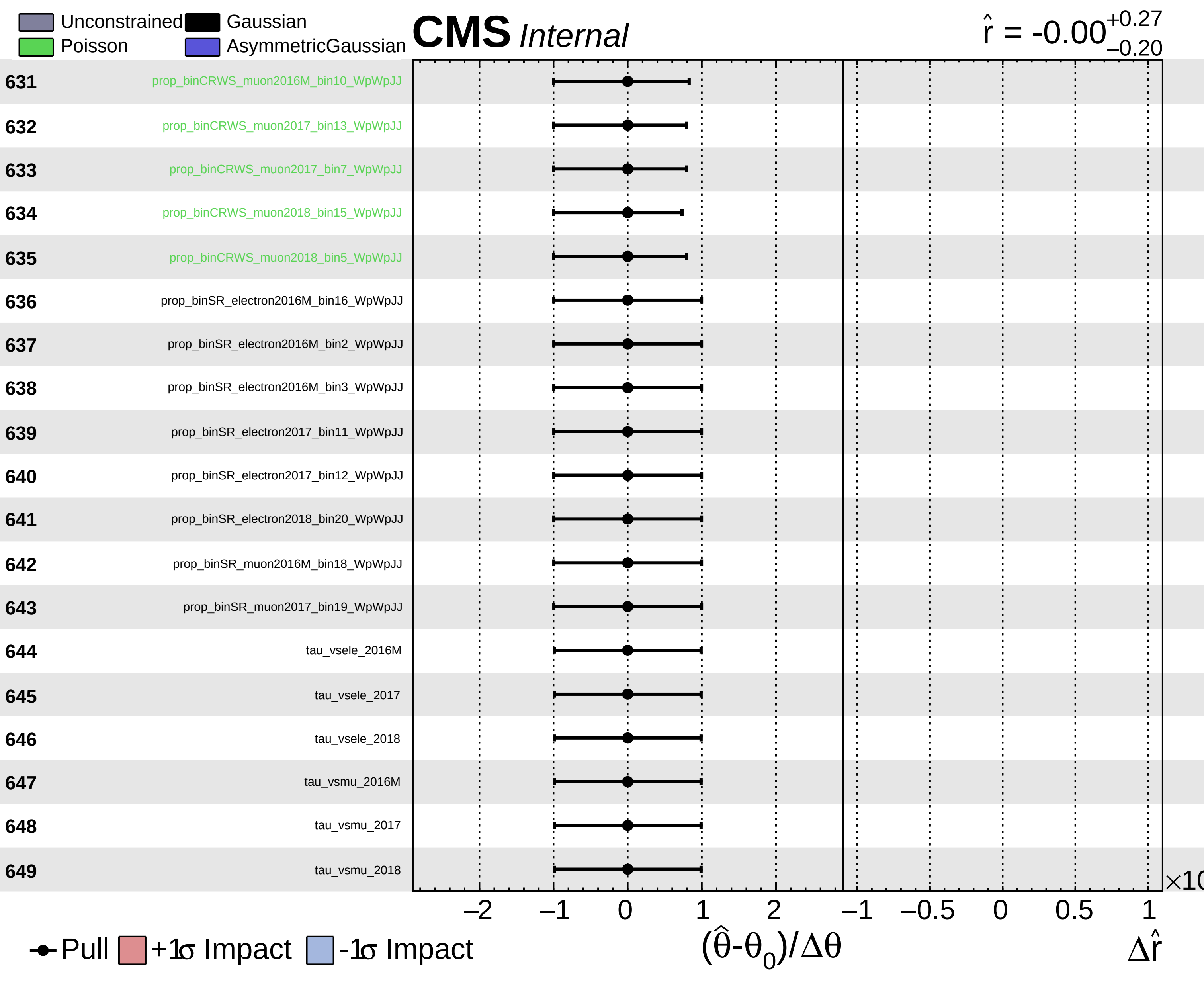

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
541 prop_binCRWS_muon2018_bin15_TVX
542 prop_binCRWS_muon2018_bin13_TVX
543 prop_binSR_electron2016M_bin2_TVX
544 prop_binCRWS_electron2017_bin13_TVX
545 prop_binSR_electron2016M_bin3_Triboson
546 prop_binSR_electron2016M_bin2_Triboson
547 prop_binCRWS_muon2017_bin7_VG
548 prop_binCRWS_muon2016M_bin10_TVX
549 prop_binSR_muon2017_bin6_ZZtoLep
550 prop_binCRWS_muon2018_bin13_Triboson
551 prop_binCRWS_muon2017_bin12_TVX
552 prop_binCRWS_electron2018_bin8_Triboson
553 prop_binCRWS_electron2016M_bin5_WrongSign
554 prop_binCRWS_electron2016M_bin9_Triboson
555 prop_binCRWS_electron2018_bin16_Triboson
556 prop_binCRWS_electron2016M_bin10_VG
557 prop_binCRWS_electron2016M_bin5_Triboson
558 prop_binCRWS_muon2018_bin15_Triboson
559 prop_binCRWS_muon2017_bin7_TVX
560 prop_binCRWS_muon2017_bin13_TVX
561 prop_binCRWS_electron2016M_bin5_WZ
562 prop_binCRWS_muon2017_bin7_Triboson
563 prop_binCRWS_muon2016M_bin10_Triboson
564 prop_binCRWS_electron2016M_bin12_TVX
565 prop_binCRWS_muon2017_bin14_Triboson
566 prop_binCRWS_electron2018_bin6_Triboson
567 prop_binCRWS_electron2016M_bin10_TVX
568 prop_binCRWS_muon2016M_bin10_VG
569 prop_binCRWS_muon2017_bin14_WrongSign
570 prop_binCRWS_muon2018_bin5_Triboson
−2 −1 0 1 2
−2 0 2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
571 FES
572 prop_binCRWS_electron2016M_bin10_ZZtoLep
573 prop_binCRWS_electron2016M_bin10_Triboson
574 prop_binCRWS_muon2017_bin13_Triboson
575 prop_binCRWS_electron2017_bin14_Triboson
576 prop_binCRWS_electron2018_bin12_TVX
577 prop_binCRWS_muon2017_bin13_ZZtoLep
578 prop_binCRWS_muon2017_bin11_Triboson
579 prop_binCRWS_electron2017_bin13_Triboson
580 prop_binCRWS_electron2018_bin16_TVX
581 prop_binCRWS_muon2017_bin14_TVX
582 prop_binCRWS_muon2017_bin11_ZZtoLep
583 prop_binCRWS_muon2017_bin14_ZZtoLep
584 prop_binCRWS_electron2018_bin12_Triboson
585 prop_binCRWS_electron2017_bin14_ZZtoLep
586 prop_binCRWS_electron2018_bin12_ZZtoLep
587 prop_binCRWS_muon2017_bin11_TVX
588 prop_binCRWS_electron2018_bin16_ZZtoLep
589 prop_binCRWS_muon2017_bin12_ZZtoLep
590 prop_binCRWS_muon2018_bin13_ZZtoLep
591 prop_binCRWS_electron2016M_bin5_ZZtoLep
592 prop_binCRWS_electron2016M_bin12_Triboson
593 prop_binCRWS_electron2018_bin8_ZZtoLep
594 prop_binCRWS_electron2017_bin13_ZZtoLep
595 prop_binCRWS_muon2017_bin7_ZZtoLep
596 prop_binCRWS_electron2017_bin11_Triboson
597 prop_binCRWS_electron2018_bin6_ZZtoLep
598 prop_binCRWS_electron2017_bin11_TVX
599 prop_binCRWS_electron2018_bin6_TVX
600 prop_binCRWS_muon2017_bin12_Triboson
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.27 −0.20
601 prop_binCRWS_electron2018_bin8_TVX
602 prop_binCRWS_electron2017_bin6_Triboson
603 prop_binCRWS_electron2017_bin6_TVX
604 prop_binCRWS_muon2018_bin5_VG
605 prop_binCRWS_electron2016M_bin9_ZZtoLep
606 prop_binCRWS_electron2017_bin6_ZZtoLep
607 prop_binCRWS_muon2016M_bin10_ZZtoLep
608 prop_binCRWS_electron2016M_bin5_TVX
609 prop_binCRWS_muon2018_bin5_TVX
610 prop_binCRWS_muon2018_bin5_ZZtoLep
611 prop_binCRWS_electron2016M_bin12_ZZtoLep
612 prop_binCRWS_electron2017_bin14_TVX
613 VBS_2018
614 prop_binSR_muon2017_bin6_WpWpJJ
615 VBS_2016M
616 prop_binSR_muon2017_bin13_WpWpJJ
617 prop_binSR_electron2017_bin19_WpWpJJ
618 prop_binCRWS_muon2017_bin12_WpWpJJ
619 prop_binSR_electron2017_bin14_WpWpJJ
620 VBS_2017
621 prop_binCRWS_electron2016M_bin10_WpWpJJ
622 prop_binCRWS_electron2016M_bin12_WpWpJJ
623 prop_binCRWS_electron2016M_bin5_WpWpJJ
624 prop_binCRWS_electron2016M_bin9_WpWpJJ
625 prop_binCRWS_electron2017_bin13_WpWpJJ
626 prop_binCRWS_electron2017_bin14_WpWpJJ
627 prop_binCRWS_electron2017_bin6_WpWpJJ
628 prop_binCRWS_electron2018_bin16_WpWpJJ
629 prop_binCRWS_electron2018_bin6_WpWpJJ
630 prop_binCRWS_electron2018_bin8_WpWpJJ
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.27}_{-0.20}$
631 prop_binCRWS_muon2016M_bin10_WpWpJJ
632 prop_binCRWS_muon2017_bin13_WpWpJJ
633 prop_binCRWS_muon2017_bin7_WpWpJJ
634 prop_binCRWS_muon2018_bin15_WpWpJJ
635 prop_binCRWS_muon2018_bin5_WpWpJJ
636 prop_binSR_electron2016M_bin16_WpWpJJ
637 prop_binSR_electron2016M_bin2_WpWpJJ
638 prop_binSR_electron2016M_bin3_WpWpJJ
639 prop_binSR_electron2017_bin11_WpWpJJ
640 prop_binSR_electron2017_bin12_WpWpJJ
641 prop_binSR_electron2018_bin20_WpWpJJ
642 prop_binSR_muon2016M_bin18_WpWpJJ
643 prop_binSR_muon2017_bin19_WpWpJJ
644 tau_vsele_2016M
645 tau_vsele_2017
646 tau_vsele_2018
647 tau_vsmu_2016M
648 tau_vsmu_2017
649 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$
Pull
+1σ Impact
-1σ Impact